

PART V

Kennedy, Johnson,
and the Technocratic
Temptation

The quest for glory is the basest thing in man; but it is just this which is also the greatest mark of his excellence. —PASCAL, *Pensées*

If the newspapers printed a despatch that the Soviet Union planned sending the first man to Hell, our federal agencies would appear the next day, crying, "We can't let them beat us to it!" —HYMAN RICKOVER, 1959

The exploration of space will go ahead whether we join in it or not. . . . We choose to go to the Moon in this decade, and do all the other things, not because they are easy, but because they are hard. —JOHN F. KENNEDY, 1962

Science and technology are making the problems of today irrelevant in the long run, because our economy can grow to meet each new charge placed upon it. . . . This is the basic miracle of modern technology. . . . It is a magic wand that gives us what we desire. —ADLAI E. STEVENSON II, 1965

Oh, you may leave here for four days in space/ But when you return it's the same old place/ . . . Hate your next door neighbor, but don't forget to say grace. —STEVE BARRY and P. F. SLOAN, "Eve of Destruction," 1965

THE maître d' of Lochober's, a Boston institution, was a longtime rocketry buff. On slow evenings Freddy would place an empty whiskey bottle on the bar, stick a pin through a straw crosswise, set the straw afire and gently lower the flame into the bottle. The pin rested on the lip while the alcoholic fumes expanded inside. Freddy then counted down dramatically to the "pop" that sent the straw shooting toward the ceiling.

Brothers Jack and Bobby Kennedy closed the Men's Clam and Oyster Bar at Lochober's more than once during the 1950s. One night Freddy introduced them to another habitué, Charles Stark Draper of MIT, hoping to persuade the young politicians that rocketry was anything but frivolous. Instead, the Kennedys heatedly dismissed the whole business. Even after *Sputnik I*, Senator Kennedy "could not be convinced that all rockets were not a waste of money, and space navigation even worse."¹

Kennedy was not defending a considered opinion; more likely, he just enjoyed the role of hard-headed skeptic. If so, he was no different from millions of intelligent Americans who supported science and discovery

but as late as 1957 still shied from anything as outrageous as men on the moon. In 1960 Kennedy was pleased to campaign on the missile gap, but of all the issues he would face as President, he "probably knew and understood least about space."² Just months later, Kennedy stood before a joint session of Congress to ask Americans to bear the burdens and costs of rocketing men to the moon within six to eight years. This extraordinary commitment, proposed on May 25, 1961, was his most historic act—but the confession of faith in spaceflight comprised only a fifth of that Special Message to the Congress on Urgent National Needs. The moon appeal, like Ike's Farewell Address, must be studied in context.

Kennedy called it a second State of the Union Address. The tradition of annual reports had been broken in extraordinary times, he said, and these were such times. Since 1941 threats to freedom had been primarily military, but now "the great battlefield for the defense and expansion of freedom today is the whole southern half of the globe—Asia, Latin America, Africa, and the Middle East—the lands of the rising peoples." Their revolution was the greatest in history, but the adversaries of freedom were "seeking to ride the crest of its wave—to capture it for themselves." They fired no missiles but sent arms, agitators, aid, technicians, and propaganda to every troubled area. He cited Vietnam, where 4,000 civil servants had been murdered the previous year. The struggle to preserve freedom in these nations was a "contest of will and purpose as well as force and violence—a battle for minds and souls as well as lives and territories. And, in that contest, we cannot stand aside."³

Such was the preamble to varied proposals the President had come to set before the Congress. The first was stimulation of the economy with an "affirmative anti-recession program" lest "we handicap our effort to compete abroad and to achieve full recovery at home." Second, the United States must foster global progress, for "the most skillful counter-guerrilla efforts cannot succeed where the local population is too caught up in its own misery to be concerned about the advance of communism." This justified Kennedy's Act for International Development (AID) and a quarter-billion dollar contingency fund for foreign aid.

"All that I have said," the President continued, "makes it clear that we are engaged in a world-wide struggle in which we bear a heavy burden to preserve and promote the ideas that we share with all mankind, or have alien ideas forced upon them [*sic*]." This justified expansion of the U.S. Information Agency (USIA). The United States must also give all necessary aid to local forces with the will and capacity to cope with attack, subversion, insurrection, or guerrilla warfare—the \$1.6 billion already requested for military assistance would not suffice. American military strength was also insufficient. Fourth, therefore, in the list of initiatives was reinforcement of the army and marines to provide "flexibility" below the threshold of nuclear war. Fifth on the list was civil defense, in case the nuclear-armed adversary took leave of his senses,

and a tripling of the budget for fallout shelters and other measures. This in turn led to hopes for disarmament and Kennedy's call for an Arms Control and Disarmament Agency (ACDA).

"Finally," announced the President,

if we are to win the battle that is now going on around the world between freedom and tyranny, the dramatic achievements in space which occurred in recent weeks [Gagarin and Shepherd] should have made clear to us all, as did the Sputnik in 1957, the impact of this adventure on the minds of men everywhere, who are attempting to make a determination on which road they should take. . . . Now it is time to take longer strides—time for a great new American enterprise—time for this nation to take a clearly leading role in space achievement, which in many ways may hold the key to our future on earth.

The Soviet lead in boosters promised them

still more impressive successes, but we nevertheless are required to make new efforts on our own. For while we cannot guarantee we shall one day be first, we can guarantee that any failure to make this effort will make us last. We take an additional risk by making it in full view of the world, but as shown by the feat of astronaut Shepherd, this very risk enhances our stature when we are successful.

This was not just a race: "whatever mankind must undertake, free men must fully share."

Kennedy then stated his belief that the United States "should commit itself to achieving the goal, before this decade is out, of landing a man on the moon and returning him safely to earth. No single space project in this period will be more impressive to mankind, or more important for the long-range exploration of space; and none will be so difficult or expensive to accomplish." A moon landing would demand sacrifice, discipline, and organization: the nation could no longer afford work stoppages, inflated costs, wasteful interagency rivalries, or high turnover of key personnel. "New objectives and new money cannot solve these problems. They could, in fact, aggravate them further unless every scientist, every engineer, every technician, contractor, and civil servant gives his personal pledge that this nation will move forward, with the full speed of freedom, in the exciting adventure of space."

Then, suddenly, Kennedy stopped trumpeting the charge and prepared a retreat. He only "believed" that the nation should go to the moon; he warned that "this is a judgment which the members of Congress must finally make"; then "this is a choice which this country must make" and "a decision that we make as a nation"; then "I think every citizen of this country as well as the members of the Congress should consider the matter carefully"; then "you must decide yourselves" and "whether you finally decide in the way that I have decided or not, that your judgment—

as my judgment—is reached on what is in the best interests of our country.”

Here were echoes of 1946 and the atomic energy debate. How could the Congress and people judge the worth of an exercise in technological futurism? How was this an “American” enterprise—the fixing of five year plans and national mobilization under federal bureaucracies? But Kennedy did not ask for judgment on the means, only on the goal. The context of the exhortation made the goal clear: to win a battle of image making in a total Cold War. In conclusion, the President praised the willingness of the American people to pay the price, to share resources, to join the Peace Corps or armed forces, to keep physically fit, to pay higher postal rates, to show friendship with foreign students that they might return home “with an image of America—and I want that image, and I know you do, to be affirmative and positive. . . .”

Daniel Boorstin, paraphrasing Oscar Wilde, wrote in 1962 that when the gods wish to punish us, they make us believe our own advertising.⁴ The French pioneers of semiotics said the same, observing how the human use of symbols—words—can supplant the very things the symbols signify, to the point where the message conveyed by a symbol can shift 180 degrees over time. When a person, company, or nation sets out to sell its image rather than reality, the image making gradually absorbs the reality: values become pseudo-values and events staged to convey the image become self-fulfilling prophecies that impress, in the end, only those whose consciousness has been conditioned to respond to the false and not to the original.⁵ The selling of the United States grew up with the Cold War. As early as 1955 press magnate William Randolph Hearst telegraphed the following to his editors, with a copy to the White House:

Kruschev [*sic*] who unquestionably the boss at present, made clear they still hope achieve communist domination world, but they want confine struggle measures short of war. That is meaning of competitive coexistence. It is that battle we must prepare for now. I think we should prepare for it with program of initiative and enterprise. It means convincing the people of Russia, China, India, as well as Europe, that our system is the best.⁶

The concomitant arrival of Sputnik and the Third World generalized the problem of the American image. The Soviet challenge and European colonial heritage made it vital for the United States to present an image of progressive anticolonialism. Perhaps, too, Americans just want to be loved. In any case, the policy or prideful want became a high priority in the context of total Cold War. But this meant the extension to foreign policy of a decadence in the United States that was the subject of Boorstin's book. Thanks to the spread of mass media—the graphic revolution—images and illusions replaced ideals in our sales-oriented

society, newsmaking and public relations replaced news gathering, celebrities replaced genuine heroes, tourists replaced travelers, the imitation of reality eclipsed reality itself. In foreign policy, the very effort to publicize the wealth and style of American life was self-defeating: "We suffer abroad simply because people know America through images. While our enemies profit from the fact that they are known only, or primarily, through their ideals."⁷ Asians and Africans were more likely to grasp Communist ideals (especially when tailored, like Jesuitical Catholicism, to local conditions) than American images of dishwashers and voting booths.

Prestige as we now understand it is a recent usage. The world originated in the Latin *praestigium*, an illusion or delusion, usually rendered in the plural to denote "juggler's tricks." In French and then English the word meant deceit: a "prestigious man" was a fraud. Only recently, especially in the United States, did "prestige" acquire a favorable connotation. In electoral politics the importance of image expanded with the advent of television, and U.S. politicians understandably transferred the techniques that won votes at home to the pursuit of goodwill abroad. Boorstin cited the example of a speechwriter who asked his client what he thought about a given issue only to hear him repeat the very phrases the agent had written for him weeks before. "It was disturbing," the speechwriter said, "to hear yourself quoted to yourself by somebody else who thought it was himself speaking."⁸ Nowhere was this phenomenon so prevalent as in outer space policy. By the end of 1958 Lyndon Johnson knew a dozen catch-phrases by heart, all coined by aides seeking to create an image for their boss and a symbolism for the Space Age. Such phrases became the ammunition dump of the space lobby, then by the 1960s, conventional political wisdom.

The brief Kennedy years were those in which American space policy fell captive to the image makers. Looking for a hook on which to hang space policy, Kennedy tossed it into the same closet with all the other policies bearing on the "extraordinary challenge" to freedom. The Kennedy call to arms amounted to a plea that Americans, while retaining their free institutions, bow to a far more pervasive mobilization by government, in the name of progress. The Apollo moon program was at that time the greatest open-ended peacetime commitment by Congress in history, the Kennedy missile program was the greatest peacetime military buildup, and McNamara's imposition of stringent management on the Pentagon and the Kennedy economic program of Keynesian fiscal policy, "pump-priming" and "fine tuning" of the economy on the assumption that private behavior was susceptible to political control, all expressed a growing technocratic mentality. The justification, at least at first, was the need to compete with the Communist bloc, in nuclear arms, conventional arms, foreign aid, economic growth, space, and propaganda. The result was an American-style mobilization that was one step away from

srashchivanie (interdependence) between the public and private spheres. That last step would come when the New Frontier gave way to the Great Society.

For the commitment to go to the moon did more than accelerate existing trends in space. It served as a bridge over which technocratic methods passed from the military to the civilian realm in the United States, to political problems at home as well as abroad. Sharp disagreements arose over the goals that government ought to pursue, but by 1964 little dissent remained over the methods. Under the impact of total Cold War, with the space program serving as lever, Left and Right, dove and hawk succumbed to the technocratic temptation.

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CHAPTER 15

Destination Moon

"The generation that fought the war"—these were the Kennedy men. Convinced of their brilliance in comparison to the men who surrounded Ike, they extolled vigor, intellection, and movement. Behind the clichés about "company commanders" replacing the generals was the truth that World War II was the formative experience of their lives. They remembered the bitter fruits of appeasement, but above all the way war had galvanized science, industry, and government, and showed what Americans could do with technology, the proper leadership, and the inspiration of a mighty cause.¹ Kennedy commanded a PT boat, John Kenneth Galbraith helped to draft the strategic bombing survey, Walt Rostow picked targets for armadas of B-17s and B-25s, Robert McNamara and his "whiz kids" supervised development of the huge B-29. Except for the latter, they knew little of design and production, but that was the point. Scientists and engineers, while they welcome the financial rewards flowing from political promotion of technology, are less likely to oversell it as a cure for all ills; they know their limits. Rather, it is the lawyer, economist, journalist, or politician who is most susceptible to technocratic temptation.

Or the company commander. Eisenhower had little faith in centralized management of power outside the military arena. But of these Best and Brightest, David Halberstam wrote, "if there was anything that bound the men, their followers, and their subordinates together, it was the belief that sheer intelligence and rationality could answer and solve anything."² To set goals for the nation and devise methods for their achievement under state direction: this was the approach to public policy that captured university faculties and foundations in the late 1950s. Political scientists like James MacGregor Burns despaired of Eisenhower's passivity and wanted an activist presidency. FDR was the favored model. It seemed obvious that the United States could do better, that official reticence only perpetuated the ills of society, that power was not corrupting but a tool to be used for good. To set goals for the people, to assume command as the most intelligent and inspired citizens, this was simply leadership, no less.

The environment changed as well as personnel. Technological revolution was abroad in the world, and limits to action retreated beyond the horizon. In such a historical conjuncture Eisenhower's philosophy seemed not only obsolete but immoral, while a mobilized United States knew no limits. Kennedy said as much in his inaugural address: "The world is different now. For man holds in his mortal hands the power to abolish all forms of human poverty and all forms of human life. . . . Let the word go forth from this time and place, to friend and foe alike, that the torch has passed to a new generation of Americans . . . we shall pay any price, bear any burden. . . ." ³

How different from Ike's words eight years before, when he hoped to liquidate the Korean War, slash defense spending, end regulation. Yet what expectations lay behind the eloquence? Within months Kennedy fired toward Capitol Hill a salvo of new spending measures and within a year the largest tax cut in recent memory. Only two assumptions could underpin such actions. Either a great surplus of wealth had built up in the 1950s (giving the lie to Democratic claims that the country had been "standing still" and "living off capital") or else explosive growth was expected in the coming decade sufficient to cover "any price, any burden." How could this be? Two to 3 percent growth would not yield "the revenues required for the welfare goals he had articulated, for the expanded infrastructure the cities required, or for the national security goals [Kennedy] had set," wrote Rostow. In other words, where traditional economics dictated the setting of state spending according to the ability of the economy to bear it, the new economics dictated stimulation of the economy to the point where it could sustain the desired level of spending. Kennedy and Walter Heller, chairman of the Council of Economic Advisors, forged a consensus in favor of "lifting the level of employment and the rate of growth by unbalancing the federal budget, grossly if necessary." Investment would be encouraged, wages and prices restrained by "jawboning" and a new "social contract," and business convinced that "a large, purposeful deficit" was sound policy. ⁴

Technology did not emerge from the start as a primary tool for enforced growth. But the new dogma that federal spending was beneficial to the economy and the "pay any price" mentality conditioned the Kennedy team to think of space exploration in terms of ends (were they desirable?) rather than means (can we afford it?). When a new Soviet spectacular, Third World setbacks, and the energetic advocacy of Vice President Johnson combined to force a decision on ends in space, the outcome was assured. It was Destination Moon.

For all their "space gap" talk the Kennedy men had little notion of what to do with the space program after election day. Twice in December 1960 the President-elect met with one teammate who did, and Kennedy gave to LBJ the responsibility for space in the new administration. His

vehicle for doing so would be the National Aeronautics and Space Council, created by Johnson in the space act but hardly active since. Its first meeting had taken up important matters such as the transfer of JPL to NASA. But in its second meeting Eisenhower dozed off during a discussion of the NASA logo. The council met seven more times, with the President usually in attendance. But he and Glennan resolved to abolish it as early as 1959, only to be blocked by the Senate.⁵ Now Kennedy and his aide Theodore Sorensen decided to vest the chairmanship of the body in the vice presidency and provide for an executive secretariat. That post would be filled by Edward C. Welsh, economist, former aide to Symington in the 1956 air power hearings and contributor to JFK's speeches on space during the campaign.⁶

Johnson also grasped the threads of space policy in the Senate, where he chose his successor as chairman of the Space Committee. It was Robert Kerr (D., Okla.), an oil millionaire who knew little about space but was a cagy ally. (Kerr once boasted, "I represent myself first, the state of Oklahoma second, and the people of the United States third—and don't you forget it!")⁷ With these institutional pieces in place, Johnson set out, just as in 1957, to marshal the information and influence needed to push through an accelerated space program.

What was to be done? Eisenhower had reluctantly granted the importance of prestige in space against the judgment of his scientists. Kennedy's scientific advisers also felt that prestige was overemphasized. Jerome Wiesner of MIT headed Kennedy's Ad Hoc Committee for Space and concluded, with support from the likes of Trevor Gardner and Edwin Land, that science was the only portion of the U.S. space effort free of severe defects. Their report denounced Project Mercury, which only "strengthened the popular belief that man in space is the most important aim of our non-military space effort," and held that "a crash program aimed at placing a man into orbit at the earliest possible time cannot be justified solely on scientific or technical grounds." The committee urged Kennedy to stop advertising Mercury lest he associate himself with a possible failure or even death of an astronaut. Instead, the U.S. government should concentrate on scientific and commercial applications such as communications satellites.⁸ This Wiesner Report comprised a scientists' critique that would echo until the moon landing and beyond.

Kennedy found the report "highly informative" and promptly named Wiesner his Special Assistant for Science and Technology—then he set the report aside. "I don't think anyone is suggesting that their views are necessarily in every case the right views."⁹ The admonition that seemed to affect the new President the most was that concerning Mercury—exploding rockets, dead astronauts, lost races—that is, not that manned spaceflight was wasteful or misguided, but that it might be a public relations failure. In his news conference of February 8, 1961, Kennedy

"They Went Thataway"



From *Straight Herbblock* (Simon & Schuster, 1964). Originally appeared in the *Washington Post*, December 30, 1960.

demurred on the race for man in space, placing safety above the desire to "gain some additional prestige."¹⁰

Hence the first months of the new administration showed hesitancy about space rather than bold forays into this new frontier. Kennedy was learning, Johnson preparing his ground. The PSAC opinion was already on the table, but ran counter to the visceral enthusiasm of Johnson and the Congress. The only actor missing from the scene was the new NASA administrator.

Glennan resigned in December, and Dryden, the apolitical expert, was asked to stay on as acting administrator through the transition. But what should the new man be like: a low-profile technician, a businessman, ex-general, university president, political wheeler-dealer? The choice would be a function of what the NASA chief would be asked to do.

Reflecting its early confusion on this score, the transition team interviewed two dozen candidates, including James Gavin. He was an attractive choice, since he understood the R & D cycle as well as anyone, supported NASA despite his views on the military importance of the "space theater," criticized Eisenhower, and expected space technology to spark an economic revolution.¹¹ But Gavin either turned down the job or was scratched as a military man. Frustrated, Kennedy tapped Johnson to fill the vacancy, Johnson consulted Kerr, and the latter touted his business partner James E. Webb—the same man who had served Truman as Director of the Budget and axed the early ICBM and satellite programs!

Webb was fifty-four years away from the rural North Carolina of his birth when he took control of the civilian space program. Trained as a lawyer in the capital, Webb became a reserve pilot in the marines and an officer in Sperry Gyroscope Company in the 1930s. He joined the Truman administration in 1946. During the Republican ascendancy, Webb had made his fortune with Kerr-McGee Oil, sat on the board of McDonnell Aviation, and given considerable time to public service. This included leading roles in the Municipal Manpower Commission devoted to urban problems, the Meridan House Foundation, a center for foreigners in the United States, and Educational Services, Inc., in which Webb collaborated on a high school physics text to meet the needs of the Space Age. In sum, he was steeped in the post-Sputnik ethic of government activism, prestige, and scientific mobilization. But when Webb arrived in Washington on a weekend late in January 1961, he told Dryden, "Hugh, I don't think this job is for me." Dryden replied, "I agree with you. I don't think it is either." Webb sent friend Frank Pace to appeal to LBJ, but he got "chased out of the office." He then saw Philip Graham, publisher of the *Washington Post*: "Phil, I've got to get out of this, can't you help me?" No, said Graham, the only man who could was Clark Clifford of the Kennedy transition team. But Clifford had also recommended Webb: "I'm not going to help you get out of it." So on Monday morning Webb reported to the Oval Office. Kennedy explained that he did not want a scientist at NASA but "someone who understands policy . . . great issues of national and international policy." Slightly mollified, Webb accepted: "I've never said no to any President who asked me to do things."¹²

In the words of Abe Zarem, president of Electro-Optical Systems, a NASA administrator had to be

... a missionary, an evangelist, with a keen sense of our national rendezvous with destiny ... an efficient manager ... suave, a man of exceptional social manners, particularly for briefing Congress ... able to understand human beings to keep in effective operation people of extremely diverse personalities ... understand the relationship between scientific knowledge and industrial might ... know generals and admirals ... know the "spaghetti bowl," the Pentagon,

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how it works and how to get around it . . . understand the workings of the Budget Bureau.¹³

James Webb was such an extraordinary man. But there would have been no point in placing him in charge of a space program limited to small-scale science or mortgaged to the military. In fact, Webb took office on February 14 anxious "to make unmistakably clear our support for the manned spaceflight program. . . ."¹⁴

The new team was in place. Lawyers, politicians, businessmen, academics, they were confident of their ability to manage a vastly expanded program of civilian command technology. But before the new team could even put the issue of NASA's future before the new President, they had to fight off another challenge to the *raison d'être* of NASA itself. The outcome of this skirmish, like the philosophy of the new administration and the choices of Johnson, Kerr, Welsh, and Webb, narrowed the possible futures of the American space program and pointed it, incredibly, toward the moon.

The *melée* over control of space R & D and operations after Sputnik left only two standards flying, those of NASA and the USAF. Military, and some civilian, critics still questioned the wisdom of a divided program. If military control was obnoxious, then let NASA do everything, but unify the program somehow!¹⁵ USAF space managers considered the verdict of 1958-59 irrational, unjust, and possibly dangerous. American rocketry grew up in the services. Farsighted officers had pleaded for years for the funds to launch the Space Age. But as soon as Sputnik had vindicated them, the government said, "OK, you were right. Now take all that you have done and hand it over to this new, civilian group." So the USAF space cadets waited, assuming the battle lost but the war still on, until the day when NASA might fade back into the status of the old NACA. This did not mean that the USAF did not cooperate; rather it must help NASA push space technology forward against the day when it might share in the spoils.¹⁶ In the meantime, its skillful PR apparatus advertised USAF experience and prowess in space, kept the problems of a divided program before the public, and declared that "peaceful uses of space" were best ensured by a strong U.S. military presence in orbit.¹⁷

The presidential campaign, with its promise of change in the midst of "missile and space gap" mania, seemed an opportunity for the USAF to recoup. In October 1960, General Schriever established an Air Force Space Study Committee under Trevor Gardner, the man who had championed the crash program for an ICBM.¹⁸ Meanwhile, the Air Force Secretary's office and aerospace trade press publicized the military shortcomings of the current space program.¹⁹ The USAF Space Study Committee met five times over the winter and issued its top-secret Gardner Report on March 20, 1961. The first sentences revealed its

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position: "The military implications of the frequency and payload size of the Soviet space launches are a major cause of alarm for all members of the Committee. Under existing U.S. schedules, and with the present organization, it will be *three to five years* before we can duplicate the recent Soviet performance." Soviet men in space, orbital rendezvous, and lunar exploration posed an "impending military space threat" that could not be met by current space organization. Among the hurdles was "the insistence on classifying space activities as either 'military' or 'peaceful.'" The Soviets made no such distinctions, while American niceties only exposed the United States to political attacks. Thus the divided program was the worst of both worlds. The panel recommended that a new Air Force Systems Command be given the task of developing manned spaceflight, space weapons, reconnaissance systems, large boosters, space stations, and even a lunar landing by 1967-70. The U.S. military, after all, had a long history of leadership in exploration, and in any case the inhibitions against military spaceflight approached a unilateral arms moratorium. "The U.S.," lamented the report, "has a consistent record of under-reacting to the rate of Soviet technological and military progress . . . in the military space field, we have continued to under-imagine the possibilities of the future and are not yet organized to exploit them." NASA, deemed superfluous, was scarcely mentioned in the sixty-four-page report.²⁰

It probably never occurred to USAF petitioners that the new administration would embrace their ambitions, grant their military importance, and still weigh in on the side of the civilian agency. Yet the aftermath of the USAF space gambit was precisely that. Even before Gardner reported, the chairman of the House Space Committee, Overton Brooks (D., La.) sniffed the winds and preempted the assault. He told a White House conference in February that "any step-up in the [space] program must be designed to accelerate a *civilian* program of peaceful space exploration and use. . . . This is very important from the standpoint of international relations." The military had a legitimate role, but "NASA and the civilian space program badly needed a shot in the arm." A big space program would have a "pronounced and beneficial effect on America's civilian economy" as well.²¹ Three weeks later Brooks expressed to the President his serious concern about persistent rumors to the effect that radical change was about to take place in space policy in the direction of military uses.²²

While awaiting Kennedy's answer, Brooks sponsored hearings on DoD involvement in space. Undersecretary Roswell Gilpatric assured the committee that the DoD did not want to control NASA: "We have plenty of problems today. We don't need any more." When USAF General Thomas White took the stand, he deftly retreated. To be sure, he had spoken of NASA "combining with the military," but that was only a statement of possible fact, not of advocacy, and was meant to encourage

USAF commanders to cooperate, not compete, with NASA.²³ Chairman Brooks was pleased with this assurance but even more pleased with President Kennedy's reply to his letter: "It is not now, nor has it ever been, my intention to subordinate the activities in space of NASA to those of the DoD."²⁴

The USAF gambit was checked: if a major expansion in space should occur, NASA would be the beneficiary. But had the USAF been frustrated as thoroughly as it appeared? Did Generals White and Schriever really hope to persuade the politicians that a single, military-run space program best met all desiderata? No one knew more about management of space R & D than Schriever, yet even as the Gardner Committee slammed the civilian program, Schriever lectured to an engineering convention in Pittsburgh on the *divergent* needs of civil and military spaceflight. They were complementary, he said, and both must be pursued with imagination and vigor. But "the military and civilian missions . . . do not merge into a single image." The technology was essentially the same, but even this was a temporary condition. First, the military would need many more space vehicles (for surveillance, communications, etc.) than NASA, since the latter's would be exploratory in nature. Second, military spacecraft would have relatively longer lives and highly repetitive missions, while scientific spacecraft changed payloads almost with every shot. Third, military vehicles must be simple, reliable, and easy to maintain, while scientific ones would be complicated. Fourth, military missions were time-critical, while NASA, certain launch windows excepted, could choose when to fund and execute projects. Finally, military space technology required close coordination between developer and user, while in NASA programs the same team of technicians served as designers and users. All this meant that different management challenges faced the two programs. Schriever predicted that NASA and USAF efforts would diverge over time, implying that a dual space program was indeed appropriate.²⁵

If these were the professional insights of the leading USAF space executive, then what are we to make of the Gardner Report he commissioned? It appears likely that the USAF deliberately overstated its case in order to educate the new administration into USAF assumptions about the Soviet military space threat. It hoped for greater, if not total, support for a "race posture," experimentation with military applications, and military participation in manned spaceflight and big boosters. General White might have been sincere in predicting large operational missions in space for the USAF, but that would come later, as the programs diverged, not at once, when R & D was still to be done. Schriever's own contribution to the policy process bears out this interpretation. He earnestly supported an accelerated NASA program, in order to stack the "building blocks" of spaceflight.

USAF background noise did make an impression. Webb's first priority

at NASA was to cement ties with the DoD, while Secretary of Defense McNamara initialed Eisenhower's planned increase in military astronautics for FY 1962. Abraham Hyatt of NASA then proposed a division of tasks between the two agencies that granted the DoD primacy for military missiles, reconnaissance, military communications satellites, navigation, geodesy, satellite inspection and interception, and a joint role in launch vehicles and manned spaceflight.²⁶ On February 23 McNamara and Webb agreed that neither agency would initiate development of new launch vehicles without the other's consent, while large solid-fueled rockets were to be a USAF show. The DoD also retained a stake in manned flight with its X-20 Dyna-Soar and was promised the opportunity to observe and learn from Mercury.

By the end of March 1961, when Kennedy finally turned to space policy, not only the Gardner Committee but also the Space Science Board of the NAS had rebutted Wiesner and come out for a vastly expanded space program. Heretofore scientific views on the space program had been hostile to "big engineering" as opposed to research satellites. But the NAS Space Science Board, chaired by Lloyd Berkner, a close friend of Webb, recommended that *"scientific exploration of the moon and planets should be clearly stated as the ultimate objective of the U.S. space program for the foreseeable future."* It considered that "[f]rom a scientific standpoint, there seems little room for dissent that man's participation in the exploration of the Moon and planets will be essential."²⁷ The board also held (a bit beyond its competence) that "the sense of national leadership emergent from bold and imaginative U.S. space activity" pointed toward a large manned program, and that "man's exploration of the Moon and planets [is] potentially the greatest inspirational venture of this century and one in which the whole world can share; inherent here are great and fundamental philosophical and spiritual values which find a response in man's questing spirit and his intellectual self-realization."²⁸

Here was language to stoke the visionary, intellectual President! The scientifically sound but uninspiring caveats of the Wiesner Report fell flat by comparison. More important, the Space Science Board altered the terms of debate. Beforehand, the main conflict had been one of politicians and engineers pushing manned spaceflight for prestige, security, or big budgets, *versus* scientists and treasurers favoring unmanned flight because of greater scientific returns and much lower costs. But now a body of scientists had come out for a manned moon program, asserted its scientific value, and appealed to something more than "knowledge gained per dollar spent." Manned spaceflight could now be viewed as something over which "good scientists disagree"; the weight of purely political judgments was accordingly enhanced.

The minutes of the Space Science Board meetings, however, tell a different story. It seems that Berkner himself proposed that the board

offer an opinion on manned spaceflight, and several members who spoke in favor of it did so in hopes that it would turn nations away from weapons and war, *not* because they held it to be an efficient scientific investment. Other board members bluntly criticized manned spaceflight as misdirected. Berkner then tried to close the meeting with the observation that "this sort of negativism always appears" and that a clear-cut national decision was needed at once. Several members then disputed the wisdom of the board making any policy recommendations, but the committee finally let Berkner "pull together a statement of the Board's position." That statement was hardly representative, but Berkner assured Webb in late February that his report would support the NASA chief's recommendations on manned spaceflight.²⁹

The support was timely. Webb confronted the BoB in mid-February only to learn that "we are still pretty much in the dark as to what position the administration desires to take in the space field."³⁰ But Budget Director David Bell urged Webb to do a quick review (the delay in Webb's appointment had caused NASA's budget process to slip) and recommend any changes before supplemental requests went to Congress in late March. When Webb canvassed NASA opinion, he learned that staff recommendations included acceleration of the big Saturn booster, an even bigger Nova, and the start to Project Apollo that Ike had denied. The Manned Lunar Landing Task Force under George Low even thought the NASA plan for a lunar landing "after 1970" too conservative: it could and should be done before 1970.³¹ Technical considerations were less important in selling the program, however, than political ones. "It is our responsibility," said Webb on March 17, "... to assess the worthwhile social objectives of our space program and to study our space effort in the context of our broad national and international goals."³²

The same day NASA made its first pitch to the BoB, asking for a 30 percent increase in the last Eisenhower budget. The case to accelerate, wrote Bell, "was well presented by Mr. Webb and his associates." But budget directors are professional skeptics. Seconded by Shapley, Bell questioned whether the United States should run races it might lose anyway, whether there were not better (and cheaper) ways of enhancing prestige, and whether "the total magnitude of present and projected expenditures in the space area may be way out of line with the real values of the benefits...."³³ Bell wrote this to the President, then told Dryden not to expect rapid action as Kennedy had other problems to worry about. Dryden retorted: "You may not feel he has the time, but whether he likes it or not he is going to have to consider it. Events will force this."³⁴

Webb got his first crack at the President on March 20. His presentation was a prototype of the technocratic argument he would make over and over again in years to come. He began by reminding the President of the effects of Soviet "firsts." To be sure, the Republicans had funded

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extensive scientific research in space (NASA's Robert Jastrow had just reported that the United States *led* the Soviets in every area of space science),³⁵ but they left the United States no room for initiative, the key to which was big boosters. Furthermore, the DoD benefited from NASA programs, while in foreign policy, the civilian space program was a positive force: "We feel there is no better means to reinforce our old alliances and build new ones. . . ." But future prospects were even greater, when

it will be possible through new technology to bring about whole new areas of international cooperation in meteorology and communications. . . . The extent to which we are leaders in space science and technology will in large measure determine the extent to which we, as a nation, pioneering on a new frontier, will be in a position to develop the emerging world forces and make it the basis for new concepts and applications in education, communications, and transportation, looking toward more viable political, social, and economic systems for nations willing to work with us in the years ahead.³⁶

Prestige, cooperation, emerging world force, viable socioeconomic systems, a new frontier—Kennedy may not have known much about space, but he knew appealing slogans. The President agreed to include \$125.7 million for rockets in his defense message to Congress of March 28. Moving Saturn ahead also kept his options open; it bought time.³⁷

Two weeks later the time ran out. Yuri Gagarin orbited the earth, and American newspapers again echoed the Kremlin's judgment of it: "a psychological victory of the first magnitude"; "new evidence of Soviet superiority"; "cost the nation heavily in prestige"; "marred the political and psychological image of the country abroad"; "neutral nations may come to believe the wave of the future is Russian."³⁸ In Congress, Space Committee members insisted that the administration, committed as it was to vigor, determine once and for all whether the United States was going to be first in space. James Fulton (R., Pa.) demanded public acknowledgment that "we are in a competitive race with Russia. . . ."³⁹ To James Webb he said, "Tell me how much money you need and this committee will authorize all you need"; to the press corps: "I am tired of coming in second best all the time." Victor Anfuso (D., N.Y.) threatened a congressional investigation: "I want to see our country mobilized to a wartime basis, because we are at war."⁴⁰ Chairman Brooks demanded that the White House do whatever was necessary to gain unequivocal leadership in space. Webb observed: "The committee is clearly in a runaway mood."⁴¹

Kennedy's initial reaction to Gagarin was not unlike that of Eisenhower after Sputnik. His congratulatory letter to Khrushchev spoke of cooperation, and he told a press conference that "while no one is more tired than I am" of the United States being second best, he hoped "to go into other areas where we can be first and which will bring more long-range

benefits to mankind."⁴² Webb, uncertain of where the President stood, lauded the broad-based scientific U.S. space program and sounded like Glennan: "The solid, onward step-by-step pace of our program is what we are more interested in than being first."⁴³

Had Kennedy and his people been genuinely contemptuous of manufactured prestige, they might have weathered this storm without major shifts of policy. But just two days after these initial utterances, Kennedy summoned Webb, Dryden, Wiesner, Sorensen, and Bell to the White House and invited journalist Hugh Sidey to observe as he played the leader intent on getting to the bottom of a crisis while others lost their heads. "Is there any place we can catch them? What can we do? Can we go around the moon before them? Can we put a man on the moon before them? What about Nova and Rover? When will Saturn be ready? Can we leapfrog them?" Webb assured him that NASA was moving ahead rapidly. Bell warned of the costs and Wiesner that "now is not the time to make mistakes." Kennedy assumed the burden: "When we know more, I can decide if it's worth it or not. If somebody can just tell me how to catch up. . . . There's nothing more important."⁴⁴

Important for what? National defense, party politics, prestige, national morale? According to historian John Logsdon, Kennedy placed space within a domestic as well as foreign context. He had suffered embarrassments in Laos and the Congo, then Gagarin and the Bay of Pigs humiliation in Cuba. Somehow the trend must be reversed. But Kennedy also had a broad domestic agenda. Would an expensive space program help or hurt him in Congress? Webb believed that a big space initiative would help Kennedy with congressional power brokers and build a basis of support for all his plans.⁴⁵ Since the space message ended up as the climax of a lengthy appeal touching on foreign and domestic programs all tied together with the Cold War ribbon, it is likely that Kennedy shared Webb's analysis.

Yet Webb himself now displayed caution. He certainly favored a big push in space, but it was he who would be responsible for it. More money for more rapid progress was one thing, but to declare a specific goal, such as those mentioned by JFK in the Oval Office, was risky. What if a moon voyage proved impossible? Or accidents should happen? This was no Lewis and Clark expedition undertaken with discretionary monies—a moon mission would mean the partial transformation of the national economy! "My own feeling," wrote Webb to Glennan on Gagarin day, "in this and many other matters facing the country at this time is that our two major organizational concepts through which the power of the Nation had been developed—the business corporation and the government agency—are going to have to be re-examined and perhaps some new invention made." Ongoing Soviet competition required the United States to "utilize every resource we have in education, communication, and transportation to build a more viable economic,

political, and social structure for the free world. . . .⁴⁶ This was technocracy, and not to be undertaken lightly. Webb was not going to get out front on it without support from the highest quarters.

Kennedy was hardly thinking in terms of national restructuring. He worried about prestige. But his advisers shared Webb's belief in the growing obsolescence of free markets, balanced budgets, or limited government. Foremost among them was LBJ. Summoned to the White House on April 19, he requested a presidential mandate to make recommendations for space. Kennedy complied:

I would like for you as Chairman of the Space Council to be in charge of making an overall survey on where we stand in space. . . . Do we have a chance of beating the Soviets by putting a laboratory into space, or by a trip around the moon, or by a rocket to land on the moon, or by a rocket to go to the moon and back with a man? Is there any other space program which promises dramatic results in which we could win . . . ?⁴⁷

The next day Kennedy announced the study to the press. The hinge was not cost—he admitted that “billions” were involved—but “whether there is any program now, regardless of its cost, which offers us hope of being pioneers in a project.” When asked if the United States should beat the Russians to the moon, he replied, “If we can get to the moon before the Russians, we should.”⁴⁸

Johnson now had carte blanche to set a goal. Rarely had a great political issue been so clear cut, but rarely had the variables been so obscure. Could giant rockets be built? Could men stand long periods of weightlessness? Could orbital rendezvous be mastered? Was the lunar surface suitable for soft landing? What were the Soviets up to? Johnson consulted NASA, which saw a chance of beating the Soviets to manned circumlunar flight or a lunar landing, but at the cost of at least \$11.4 billion extra dollars over ten years.⁴⁹ Then he asked McNamara, three business cronies, von Braun, Schriever, and Vice Admiral John T. Hayward. They all supported a moon landing, the last two stipulating only that it not detract from military missions. Von Braun even spoke of putting all other elements in the space program “on the back burner.” One of the businessmen, Donald Cook, believed that an action must be “based on the fundamental premise that achievements in space are equated by other nations in the world with technical proficiency and industrial strength . . . and will be of fundamental importance as to which group, the East or the West, they will cast their lot. . . .”⁵⁰

Johnson sent the President a report so loaded down with assumptions that a moon landing was the inescapable conclusion: (1) the Soviets led the United States in prestige; (2) the United States had failed to marshal its superior technical resources; (3) the United States should recognize that countries tend to line up with the country they believe to be the

leader; (4) if the United States did not act, the Soviet "margin of control" would get beyond our ability to catch up; (5) even in areas where the Soviets led, the United States had to make aggressive efforts; (6) manned exploration of the moon was of great propaganda value but was essential whether or not the United States was first.⁵¹ In another context, LBJ put it more pithily: "One can predict with confidence that failure to master space means being second-best in the crucial arena of our Cold War world. In the eyes of the world, first in space means first, period; second in space is second in everything."⁵²

The persuasive vice president then worked on Webb in a "consultative meeting" packed with lunar zealots: Dryden, Welsh, the three businessmen (including Frank Stanton of CBS), Senators Kerr and Bridges. All sensed that the moment for NASA had arrived. Why did Webb, of all people, hang back? He said he wanted to be sure that NASA had enough support and really knew what it was getting into: if NASA appeared to be the initiator of an expensive, questionable project, it might be left twisting in the wind should the national mood change or delays and failures ensue. NASA must be *given* the task, asked to do this risky thing by an anxious nation. Then Webb would have the leverage, down the road, to claim the backing needed to see the agency through harder times. (Indeed, five years later, Webb wrote to then-President Johnson: "You will remember that in the sessions you had in 1961 with your advisers and Congressional leaders, I was quite reluctant to undertake the responsibility of building a transportation system to the moon and that you had to almost drive me to make the recommendation which you sent on to President Kennedy." For at that May 3, 1961, meeting, LBJ was "close to demanding that NASA recommend for Apollo.")⁵³

The next day Webb wrote LBJ that he was ready to climb on board. "I think I can say also that my main effort yesterday was to be certain you and the Senators were under no illusions whatever as to the magnitude of the problems involved in carrying out this decision and the absolute necessity, in my opinion, for a decision to back Secretary McNamara and myself to the limit. . . ." Congress and the press would ride NASA "like two packs of hounds." He must be sure of the President and the Vice-President. Thrice he insisted, and thrice he referred to McNamara.⁵⁴

Why McNamara? Why did the Secretary of Defense endorse a NASA moon program? He knew the importance of prestige and of a "peaceful, civilian" space effort. But the value of a moon mission was not self-evident. McNamara's internal studies in fact convinced him that some of the more "way out" space programs were ripe for the axe, and as late as March 23 he expressed preference for a "normal rate of investigation" and "stated rather emphatically that he would accord a higher priority to items included, about to be excluded, and already excluded from the Defense budget than he would to the programs in question."⁵⁵ But there

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were still larger considerations, including the very health of the aerospace industry. McNamara knew that the missile gap was dubious and believed that defense R & D was out of control and that the aerospace lobby was the main obstacle to his plans for modern cost-accounting. But the sort of cutbacks he envisioned might damage an industry that had expanded tremendously in the 1950s on the strength of jet aviation and the space boom. What was more, government financing of idle plant capacity, cost-plus contracting, and "progress payments" (or "get paid as you go") to contractors were all being phased out. These measures hit the industry just as new plant, equipment, and personnel were needed to participate in the space revolution. The result was a profit squeeze despite record sales and a *tenfold* increase in aerospace debt in the decade after 1957. Aviation was never an easy business in which to make money, but in the early Space Age the industry superimposed high financial risk upon high business risk, a classic violation of sound corporate finance.⁵⁶

The health of the aerospace industry was as much a government worry in 1961 as in 1947. Fairchild, General Dynamics, mighty Lockheed, and Douglas were already in trouble, the latter having lost over \$100 million in 1959-60. McNamara's pet reforms, including fixed-price contracting, cost-benefit analysis even on R & D programs, and insistence on high definition even of early design work all promised to trim further the narrow margin the industry trod.⁵⁷ In this context an expensive space program, rich in new technology but lying outside the reach of the USAF and his own budget, must have seemed to McNamara, on second thought, felicitous. Giving Apollo to NASA would please the aerospace lobby and Congress, while the USAF, bereft of its allies, would nurse its jealousy alone.⁵⁸

Now everyone was on board. The Saturday after Alan Shepherd's flight found NASA, DoD, and BoB delegations gathered to discuss the least mundane of political topics: going to the moon. In the morning session Webb and McNamara exchanged reports done at LBJ's behest. In the afternoon they tackled Apollo. McNamara approved it. He believed the USAF was "out of control": Apollo would help relieve military, industrial, and congressional pressure on him so that he could get on with the tough decisions on defense programs.⁵⁹ He also believed, and everyone concurred, that large space projects "reflect the capacity and will of the nation to harness its technological, economic, and managerial resources for a common goal."⁶⁰ Apollo was proposed; no one dissented.

The secret "Webb-McNamara Report" originally contained a preamble, drafted by John Rubel of the DoD, adumbrating McNamara's references to the USAF and aerospace industry.⁶¹ It was cut in favor of the habitual rhetoric justifying spaceflight on the basis of science, commerce, defense, and especially prestige. Soviet attainments, the report suggested, were the result of a program planned and executed at the national level over a long period of time, while the United States had "over-encouraged the

development of entrepreneurs and the development of new enterprises."⁶² The United States must kick its tendency to embellish its designs. "We must insist from the top down that, as the Russians say, 'the better is the enemy of the good.'" Buried in the report was this conjecture: "It is possible, of course, that the Soviet program is not actually the result of careful planning toward long-range goals. . . . Perhaps luck played an important part. . . ." But the evidence pointed "dramatically" in the other direction. "Of all the programs planned, perhaps the greatest unsurpassed prestige will accrue to the nation which first sends man to the moon and returns him safely to earth."

Kennedy first saw the report on May 8 and met with Cabinet members on the tenth. The BoB still worried about setting dates for a moon landing and projects aimed at prestige rather than technological advance per se, not to mention the costs.⁶³ Economic advisers and Secretary of Labor Arthur Goldberg even denied that the space program would stimulate the economy. But, as McGeorge Bundy recalled, "the President had pretty much made up his mind to go," and, as Wiesner recalled, when McNamara showed the President that *without* Apollo a definite oversupply of manpower would exist in the aerospace industry, "this took away all argument against the space program."⁶⁴

Johnson returned from a trip to Southeast Asia on May 24. A letter from Webb awaited him: "The President has approved the program you submitted, with very few changes, and the message will go up on Wednesday."⁶⁵

That message asked Congress to spend upward from \$20 billion on command technology for a political goal. Compared with 1946, when the Atomic Energy Act was assailed as totalitarian, or 1948, when funding for the NSF was restricted by law, or even 1958, when Ike shied from racing in space and struggled to restrict the terms of the education act, Apollo signaled a new age. The technology race that began with weaponry now extended to a civilian pursuit, held in turn to be a symbol of overall national prowess. Where the Eisenhower men doubled and tripled spending on science, education, and R & D, it was their intention to contain as far as possible the effects on traditional values and social institutions and the relationship of the public and private sectors. The men who launched Apollo came to office dissatisfied with existing state management of the national treasure and talent, and began to view the space program as a catalyst for technological revolution, social progress, and even the "restructuring of institutions" in ways that were dimly foreseen but assumed to be "progressive."

How this change occurred in so short a time is not a mystery, but rather that most vexing of historical problems, the "overdetermined event."⁶⁶ New men arrived and brought with them those ideas of the "seed time" of the 1950s. Among those ideas were the notions that the

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Herblock, May 24, 1961. Copyright 1961 by Herblock in the *Washington Post*.

Third World was the main theater of the Cold War and that in that contest prestige was as important as power. Their new ideas validated a far greater role for government in planning and executing social change. The new men also cared more for imagery and felt increasing pressure to display their control over affairs in the wake of early setbacks in foreign policy. Finally, each major figure in space policy—Kennedy, Johnson, Webb, Dryden, McNamara, Welsh, Kerr, and others—saw ways in which an accelerated space program could help them solve problems in their own shop or serve their own interests. This is not to say that they were petty; it is to say that they were technocratic, applying command technology to political problems.

As for contrary arguments, they were disposed of, one by one. Nixon himself abandoned the original Republican skepticism toward space races even before the campaign. The USAF view that the main threat was military got nowhere with the image-conscious civilians. The scientists' argument against prestige-oriented manned spaceflight was bulldozed. When the Soviets weighed in by orbiting Gagarin, and the Shepherd flight confirmed NASA's contention that the mission was feasible, all barriers came down. All, that is, except cost, and that, too, was less important in the new White House. We will probably never know precisely what was in Kennedy's mind when he decided that Americans should go to the moon. What may have tipped the balance for him and for many was the spinal chill attending the thought of leaving the moon to the Soviets. Perhaps Apollo could not be justified, but, by God, we could not *not* do it.

Of all those who contributed to the moon decision, the ones farthest in the background were the engineers of Langley and Goddard and Marshall, many of whom devoted their lives to spaceflight, designing dreams. Their reports and studies were necessary buttresses to the political arguments: they had to persuade that the thing could be done. Otherwise, they were absent. Some of their visionary talk about exploration and destiny found place in political speeches, but their efforts to stretch the minds and hearts of their fellows, to sow wonder for its own sake, got lost in their very adoption by the technocratic state. What Constantine's conversion did to the Christian church, Apollo did to spaceflight: it linked it to Caesar. The new faith might conquer the empire, but its immaculate ability to stir hearts was accordingly diminished. Of course, it could not have been otherwise.

CHAPTER 18

Big Operator: James Webb's Space Age America

Lyndon Johnson, Speaker Sam Rayburn, Congressmen Albert Thomas and Olin Teague of key committees—Texans all; Bob Kerr, James Webb—both Oklahomans. Clichéd though it be, the immense tablelands on either side of the Red River do nurture big thinkers and doers. A Walt Disney comic book of the early 1960s hit the mark when it depicted Scrooge McDuck as a self-satisfied Big Operator, investing billions to develop the oil of some generic Middle Eastern desert and then, as an afterthought, to pipe in water and turn the desert into a garden for a tribe of midget Third World ducks. Thousands of miles of gargantuan pipeline, huge earth-movers, the latest technology arrived in convoys of merchantmen, while Donald Duck and his three nephews tagged along in dumb admiration, hoping for the chance to prove they, too, had the stuff to be Big Operators.

Jim Webb, the Carolina lawyer transplanted to the Okie oil fields, was a budget-cutter for Truman and suspicious enough of creeping socialism to slash funds for the TVA. As NASA chief, he was fond of recalling that misstep to balky congressmen, for he had been quickly brought to see that it was a false economy. An accomplished administrator, he then went West and learned, in his years with Kerr-McGee Oil and McDonnell Aircraft, the virtues of thinking big. By 1960 he had also learned the potential of progressive big government from his service on the Ford Foundation's commission on urban problems, the Educational Services Inc. Commission on post-Sputnik science curricula, the board of the Oak Ridge Institute of Nuclear Studies, and the National Cancer Advisory Council. Webb was not a scientist, but a lawyer and manager who administered the power inherent in applied science. Once given the astounding task of organizing an expedition to the moon, Webb must have felt that, improbable though the whole thing was, few men were as able as himself to tackle it. He was a Big Operator.

Apollo is often dubbed a peculiarly "American" enterprise.¹ But why? Great treks and explorations have been a part of many nations' history;

obsession with prestige has afflicted most Great Powers, not to mention big management of resources by government. The uniquely American flavor of Apollo, rather, smacks of mesquite: not just the big capitalism of Scrooge McDuck, or just the big government of technocratic social planners, but an amalgam of the two, infused with the spirit of the Southwest frontier. Here was limitless space, limitless opportunity, limitless challenge, a land where giants and giantism could live without trampling the liberty of the little. A Johnson or a Webb did not see a conflict between technocracy and freedom, between the expansionist state and the striving individual. In America's vastnesses they could coexist and feed on each other. The activist state fulfilled the individual through education, welfare, incentives, new technology; the individual, in turn, lent his ambition and skills to the collective effort. Apollo never lacked for analogies, but to the Big Operators who made it, the moon program was less like the voyages of Columbus or even the privately built railroads than it was like the Panama Canal, the TVA, or the magnificent interstate highway system. Like these earlier mobilizations of men, machines, and mortar under federal direction, Apollo would open up new realms for the individual in stimulation of the economy and elevation of the human spirit. What was more, the space program grew on its own movers and shakers until it outgrew the space race itself and seemed a model for a society without limits, an ebullient and liberal technocracy—not Space Age communism, but Space Age America.

Webb inherited a small federal agency composed for the most part of idiosyncratic scientists and engineers—brilliant or competent or merely eccentric. His total manpower in early 1961 was 6,000. Thanks to Apollo, the agency grew tenfold, but it still contained only a tenth of the personnel working on the space program. The other 90 percent were in private firms under contract. They doubled each year to reach a peak of 411,000 in 1965. NASA appropriations rode the same graph: less than a billion dollars in FY 1961, they doubled and redoubled to reach \$5.1 billion in 1964.² But perhaps the immediate 1961 shock was the hardest to absorb, for then NASA received an overall increase of 61 percent in Webb's first fourteen weeks on the job.

The flux inherent in such growth reinforced other currents of instability: about how to get to the moon, how to make sound technical decisions in advance of hardware design, how to enumerate the tasks and steps (literally millions) to be trod from the first brainstorming sessions to the landing on the moon, how to divide up and coordinate the labor of NASA's several centers, contractors, subcontractors, and university labs. Glennan had opted for a small in-house capacity and reliance instead on the aerospace firms. Webb seconded this motion and thus gave the contractor system a permanent lease on life. Glennan considered it a lesser evil, but Webb made a virtue of it. NASA was to be a model of

integration, a team effort drawing confidently on government, industry, and university. But the politico-philosophical justifications came later. At first, Webb had merely to get on with the job.

Two weeks after the May 25 call to arms, NASA organizational experts recommended reforms to accommodate the lunar mission. A manned spaceflight center must be established, ably staffed, and coordinated with the whole—it would, after all, absorb more funds than any other center except Marshall, the home of von Braun's rocket team. Liaison among the centers and headquarters must be tightened, for no more would NASA serve up a smorgasbord of projects each associated with one center. Everybody must have a piece of Apollo, hence center directors should report not to the program directors, whose interests were parochial, but directly to Associate Administrator Robert Seamans, whose new programming office could then oversee the entire budget and promote functional integration among all teams in the field. But internal organization paled in comparison to the national. The immediate and primary concern, according to the June study, was NASA's dependence on industry, the DoD, and the universities.³

Heretofore the manned program (Mercury) had resided in the Space Task Group at Langley. After the acceleration NASA was authorized to construct a massive Manned Spaceflight Center for the tasks of managing spacecraft R & D, training astronauts, and directing flight operations. In late 1961 the Army Corps of Engineers broke ground near Rice University in Houston. Also that summer Cape Canaveral was chosen as permanent launch site, and 80,000 more acres acquired on Merritt Island, Florida, astride the Banana River. Moon rockets would be so large that they could only be assembled near the point of launch. But the Florida coast was subject to heavy rains and salt air. So in addition to the gargantuan gantries and creeping transporters to move the Saturns to the pad, there arose on the cape the largest enclosed space in the world, the Vehicle Assembly Building. In September NASA elected to renovate an unused government plant near New Orleans for use by Saturn contractors, and so the Michoud, Louisiana, facility was born. But engines had to be test fired. The Marshall center at Huntsville had neither big enough test stands nor the remoteness from populated areas to accommodate the clusters of F-1 engines that would roar to life in a few years. Hence the Mississippi Test Facility, near Michoud at Bay St. Louis.

By the end of 1961, when Congress was still in a "runaway mood," when Johnson ran the NASC, Webb ran NASA, Kerr the Senate Space Committee, Overton Brooks and Olin Teague (D., Tex.) the House Space Committee, and Albert Thomas (D., Tex.) the Appropriations Committee, the Apollo complex took the shape of a crescent moon running around the Gulf of Mexico from Texas to Florida. There were sound reasons for it: year-round warm weather, deep-water transport for the big rockets (imagine the Soviet difficulties in moving big rockets on rail beds across

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soft country), and contiguity. But the political aspect is undeniable. Apollo was the largest single civilian project in history, and the rolling of the pork barrel began the instant Kennedy approved the Webb-McNamara report. On May 23, 1961, when Webb wrote LBJ that the moon message "will go up on Wednesday," he got down to business without breaking for a paragraph. "Considerable interest has been expressed in this program by members of the Congress," he began, "following your consultations with them, and as I have followed up." Congressman Thomas had "made it very clear that he and [Rep.] George [E.] Brown [D., Cal., on Space Committee] were extremely interested in having Rice University make a real contribution to the effort . . . some 3800 acres of land had been set aside by Rice for an important research installation." Apollo research had to be done somewhere, and, Webb continued, "we have looked carefully at Rice, and at the possible locations near the Houston Ship Channel. . . . I believe it is going to be of great importance to develop the intellectual and other resources of the Southwest in connection with the new programs which the Government is undertaking."⁴

Texas was especially attractive, given that Lloyd Berkner, who had pressed the Space Science Board to endorse Apollo, was establishing a \$100-million research center in Dallas, while Senator Kerr and his "interested parties in the Arkansas, White, and Red River systems," were pushing to open the whole Arkansas/Oklahoma area and develop potential for Mississippi. The Dallas-Houston axis "would provide a great impetus to the intellectual and industrial base of this whole region." Webb then imagined a national complex including centers of gravity

in California, running from San Francisco down through the new University of California installation at San Diego [LaJolla], another center around Chicago with the University of Chicago as the pivot, a strong Northeastern arrangement with Harvard, M.I.T. . . . some work in the Southeast perhaps revolving round the research triangle in North Carolina (in which Charlie Jones as the ranking minority member on Thomas's Appropriations Subcommittee would have an interest), and with the Southwestern complex rounding out the situation.⁵

Ex-Administrator Glennan guessed what was afoot. When his governor asked him to help prepare a proposal to put the Manned Spacecraft Center in Ohio, Glennan laughed: "You know, I suppose that there are 25 states doing just this at the present time, and I'll lay you a year's salary that that Center is going to Houston."⁶

Similar considerations informed the choice of industrial contractors for Apollo-Saturn. In this, California congressmen took special interest, given their state's dominance in the aerospace industry. Kerr, the *eminence grise* of the Apollo acceleration, and Representative George P. Miller (D., Cal.), who assumed the chair of the House Space Committee after Brooks

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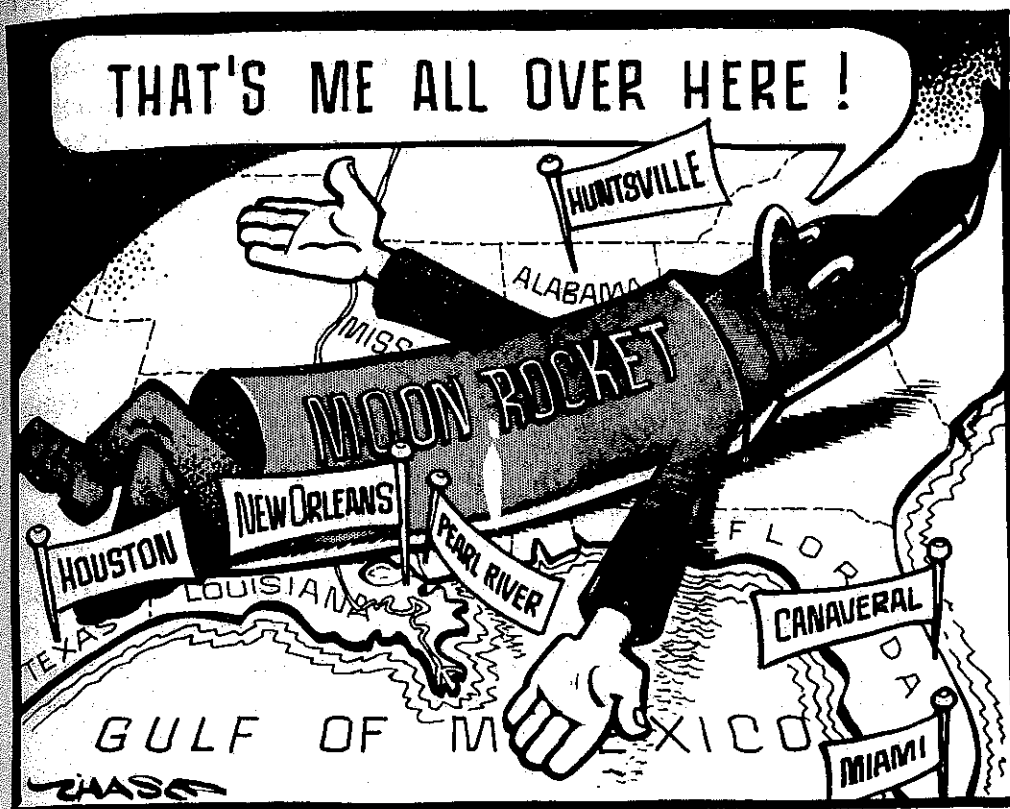
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died in September 1961, were especially interested in the North American Aviation plant at Downey, California. In July NASA briefed 300 companies on the tasks ahead and in September invited a few of the largest aerospace firms to bid on the Apollo command module. NASA then tested the five proposals according to complicated quantifications based on cost, strength of design, capabilities existing in the firm, and so on. Martin won; the contract, awarded in October, went to North American. Webb and Dryden explained that overall corporate resources and past performance tipped the balance, but muckrakers attributed it to the ostensible influence of Kerr, North American lobbyist Fred B. Black, and the shady crony of Kerr and LBJ, Bobby Baker. Kerr received promises from Black that benefits would be forthcoming for Oklahoma, Baker was granted a million-dollar concession for vending machines in North American plants, and Black, unrepentant, recalled: "North American was five times as large as Martin, so when they got the contract you had five times as many happy people as you had unhappy people. That's democracy in its finest form."⁷

That political plums ripened on NASA's bursting branches should not

surprise.* Interesting congressmen in an expensive, discretionary project is not only smart but mandatory. These plums were special, however, in two ways. First, the NASA of Apollo was a creature of the image makers—not through the fault of its own excellent engineers but through political imprimatur. The agency was supposed to provide the nation with an example of efficiency and innovation that was above politics. It was to show that free people could pull together and outperform the totalitarians at their own game. NASA was to be clean, technically perfect, and meritocratic, the bearer of a myth. What is surprising about the agency's performance through the years is that gap between image and performance was not wider than it was.⁸

The second idiosyncrasy of the NASA plums was that Johnson, Webb and perhaps others like John Stennis (D., Miss.) and Donald Fuqua (D., Fla.) viewed the political distributivism not as palm-greasing but as the economic and intellectual component of the Second Reconstruction. Of course, far more money and brokered placement of installations had flowed South through the Pentagon than through NASA. But the post-1961 NASA construction went beyond the pork barrel into the realm of social planning. Technological infusion was to call to life a New South, and the space program thus addressed several large items on the national agenda all at once.

A second management study, dating from late in 1961, sketched out new lines of authority in NASA. The central problem in Webb's eyes was how to coordinate Project Apollo's many constituent research teams and prevent them from flying off in their own directions, yet somehow provide the flexibility that centers and contractors needed to perform with imagination. The new recommendations urged that "program imbalance" be squarely faced by making the Manned Spaceflight Office a "state within a state." Second, NASA must somehow cut through the traditional layers of bureaucracy so that high-level attention could be brought to bear quickly on local problems. Third, NASA must endow individuals with the responsibility to oversee and direct all activities related to given missions *wherever* they might be taking place; that is, functional managers should be able to ignore bureaucratic "turf." If changes in policy or program were indicated, they could be taken directly to the associate administrator, to whom all center directors also reported. Fourth, Apollo-Saturn relied on systems integration like no other program save the ICBM: the administrative art of conceiving a whole, breaking it into subsystems, nursing along the R & D, testing, and evaluation of each like a cook with six dishes on the stove, and finally making sure

* The other major contracts went to Boeing (for the pentagonal array of F-1 engines for Saturn's first stage), North American again (for the second stage), and Grumman (for the lunar excursion module). Overall systems integration fell to von Braun's Marshall Spaceflight Center.

that each "interfaced" properly with all the others when time came to put the meal on the table. A design change in one subsystem often meant changes (and likely engineering headaches) in one or more other systems.⁹

All this required flexibility, coordination, oversight, cost control, and especially program monitoring. This last inspired PERT (*program evaluation and review technique*), first devised by the navy in its Polaris program. Though introduced too late to affect some NASA programs, PERT permitted computerized analysis of variables in the systematic completion of tasks. A project manager could survey all his checkpoints, hurdles, and schedules at once, and derive a realistic view of the progress of all his teams. Yet while NASA developed innovative, even experimental managerial techniques in the interest of efficiency, it also had to draft procedures for safety and quality control, clearance, and approval that met simply the highest standards imaginable. Webb was asked to do a surrealistic task, within a constricted time frame, before (in his words) three leering "packs of wolves"—Congress, press, and public.

The fruits of the 1961 management studies, implemented in November, included abolition of the old program offices and creation of four new ones for Advanced Research and Technology, Space Science, Manned Spaceflight, and Space Applications. Above them reigned Seamans, the in-house management virtuoso, Deputy Administrator Dryden, handling scientific matters and technical judgments, and finally Webb, who was left free to integrate the integrators and deal with other agencies, Congress, and the White House. Over time, Webb aimed for a vast R & D complex with management flexible enough to innovate constantly yet structured enough to run on a daily basis while Webb and his lieutenants "worked to extend NASA's influence into economic and political spheres hitherto untouched by the space program."¹⁰

Once NASA had geared up, it faced the most treacherous hurdle of all (and one that absorbed a million man-hours before it was cleared): choosing the proper mode for going to the moon. Two 1961 studies concluded that a lunar landing was attainable before the end of the decade, but one opted for direct ascent and the other for earth-orbit rendezvous (EOR). The former mode was simplest—one rocket to blast off from earth, fly directly to the moon, land, and return home. But it required an immense booster, the Saturn follow-on called Nova consisting of perhaps eight F-1 engines in the first stage and developing 21 million pounds of thrust. Nicholas Golovin and DoD's Laurence Kavanau, authors of a second study, doubted that Nova could be ready in time. They favored EOR, whereby the lunar fuel tanks would be launched into earth orbit, with the astronauts to follow in a second rocket, rendezvous, and take off for the moon in a more compact spaceship. This multiplied the chances of launch failure and involved tricky orbital

maneuvers, but the boosters would be smaller (perhaps four F-1s in the first stage), cheaper, and more likely to meet the deadline.

Throughout 1961 the Manned Spacecraft Center under Robert Gilruth, the Director of Launch Vehicles and Propulsion, Milton Rosen, and the Huntsville rocket team under von Braun pored and feuded over data favoring EOR and direct ascent. But the pacing element, clearly, was booster development. When in December 1961 the Huntsville rocketeers decided that the next step beyond *Saturn 1* was to be a five-engine first stage arrayed like the five spots on a die, Nova was out, *Saturn 5* was in, and the question then became how to get to the moon with such a rocket. Meanwhile, engineers at Langley and at Chance, Vought Inc. plumped for a third mode, lunar-orbit rendezvous (LOR), unaware that the Russian Kondratyuk had suggested it forty-five years before. The spokesman for the Langley group was John Houbolt, who took his presentation on the road seeking to persuade others that LOR was, in some respects, the best of both worlds. It needed only one launch from earth and still offered weight savings stemming from the separation of craft during the journey. Only a lunar module would descend to the moon, only a portion of it would return to dock with the command module, and only the Apollo capsule itself, with heat shield, would fall back to earth. It promised 10 to 20 percent in cost savings and the benefit of rendezvous techniques. First Houston, then Huntsville, came around to LOR and agreed to recommend Houbolt's dark horse to NASA headquarters on June 7, 1962.¹¹

The ultimate variable, of course, was not feasibility or time or cost, but safety. Which mode promised the best shot of getting astronauts back alive? EOR supporter Golovin was the reliability expert at headquarters, and he made himself obnoxious and eventually unemployed with computer-derived safety estimates damning LOR. But now PSAC entered the fray. Having opposed the moon decision and lost, Wiesner formed an Apollo Committee, hired Golovin, and led the PSAC into battle against LOR. Scientific opinion again registered displeasure with Apollo's stress on prestige and big engineering and opportunism (get us to the moon, in whatever fashion) over pragmatism (develop useful earth-orbit capability). Wiesner and von Braun even crossed swords during a presidential visit to Huntsville, tarnishing somewhat the shining patina of scientific planning that was meant to dazzle a deferential world.¹² The PSAC proved incorrigible, but NASA stuck by LOR and awarded the lunar module contract to Grumman in October 1962. When the Cuban missile crisis erupted, preoccupying the President, Wiesner resigned himself to another defeat.

The lunar mode decision completed the restructuring of NASA. Following Mercury, an interim program, Gemini, would provide NASA with experience in multimanned spaceflight, rendezvous, spacewalks, and ground control of multiple spacecraft—all the skills needed for LOR.

Many "building blocks" of spaceflight would be hewn after all.¹³ But the PSAC dissent stood—LOR was a "technological dead end" of limited future value to the space program; once current interest in lunar voyages flagged, the Saturns and Apollo craft would become extinct.¹⁴ In retrospect, one can only applaud the success of LOR. It may indeed have been the only way to reach the moon by 1970. The Soviets, by contrast, never

MERCURY-ATLAS 8 AND 9

The Mercury series of "lucky 7" capsules seemed to be pulling the United States closer in the space race. In August 1962 Wally Schirra made final preparations for a six-orbit mission. Then suddenly *Vostok III* and *IV* orbited together on August 11–12. The Mercury program office weighed (literally) the idea of adding a near-rendezvous mission (with a passive target) to the flight plan for MA-8. It meant an extra 400 pounds of hardware and fuel, on a mission that already called for relaxation of the rules on oxygen reserves and a push beyond known limits of Mercury's endurance. NASA, unlike the Soviets, refused to improvise in response to the opponent.

Another obstacle popped up on the increasingly complicated vertical frontier. The Dominic nuclear tests had created another band of radiation below the Van Allen belts, necessitating a delay. It was not until October 3, 1962, that the Atlas fired Schirra's *Sigma 7* into the standard Mercury orbit, where it spent nine hours in weightlessness, while below NASA personnel completed paperwork for approval of LOR and Soviet ships unloaded missile components in Cuban harbors. Schirra executed yaw maneuvers by hand, was loath to return to automatic controls ("the chimp configuration"), and reported the amazing detail he could make out with the naked eye: "might as well be in an airplane at 40,000 to 50,000 feet altitude."¹⁵ *Sigma 7* reentered perfectly to be picked up by the aircraft carrier *Kearsarge* and win the epithet of "textbook flight."

The final Mercury mission was a genuine capstone. The MA-9 spacecraft ("capsule" had been dropped from the lexicon) would be an advanced design suitable for a whole day in space. Gordon Cooper dubbed it *Faith 7*, expressing his "trust in God, my country, and my teammates," in retaliation for the atheistic quips of the cosmonauts. There was so much photographic gear on board when Cooper went up on May 15, 1963, that he called it "practically a flying camera." His insistence that he could make out objects as small as trucks trailing dust on a Texas highway or chimney smoke from Tibetan huts must have piqued interest in USAF planners scratching to justify a manned military presence in space.¹⁶ Cooper descended after twenty-two orbits. Again, it seemed not to bear comparison to *Vostok III*'s sixty-four orbits, or *Vostok V*'s eighty-one a month later. But it punctuated the Mercury series not with a period but an ellipsis leading to the more complicated Gemini, for which more men and resources were already mobilized—eighteen months in advance—than were employed in all of Mercury.

got to the moon, but their EOR mode blended into an evolving space station program that sustained overall progress through the 1970s.

By 1963 the U.S. civilian space program reached \$3.7 billion per year, of which almost \$2.5 billion went for R & D, 69 percent of that for manned spaceflight. That left some \$750 million for nonmilitary, unmanned R & D.¹⁷ In later years critics would demand cuts in prestige flights in favor of "practical" programs for science and applications. But in the first such reevaluation it was the other way around: Kennedy instructed Webb to report on the implications of reaccelerating Apollo to aim for a 1966 moon landing, while cutting back on other space programs. Webb replied in discouraging terms. A late 1966 landing, he wrote, "would require a crash, high-risk effort," with only a 50-50 chance of success and an additional \$1.7 billion.¹⁸ In November 1962 Kennedy raised the matter again. Perhaps the money to get Apollo over with quickly could be found in other NASA programs—the scientific satellites to study the earth, sun, planets, and stars, satellites to test new technology and applications, international cooperative programs. Webb responded in a tactful brief that preeminence in space required more than just going to the moon. To be sure, Gemini and Apollo were the natural focus of the program, but there were "many significant events by which the world will judge the competence of the U.S. in space." More important than accelerating Apollo was understanding that even the scientific programs contributed to technical progress, while technology in turn opened new vistas for space science. Nor could prestige be separated from the other things NASA did. In sum, the United States must build for activities beyond Apollo, requiring "that we pursue an adequate well-balanced space program in all areas, including those not directly related to the manned lunar landing."¹⁹

Webb won his case, but the next spring, in the midst of congressional criticism, Kennedy queried Johnson on the real value of Apollo. What were the "most salient differences" between his space program and Eisenhower's? What were the economic benefits he could point to in defense of the big space program? How might industry, government, and education suffer from continued high spending in space? Finally, to what extent could the program be reduced and not affect the Apollo schedule?²⁰

These repeated inquiries mark a turning in the U.S. space program, for once the Cold War seemed to subside, Kennedy needed to show that Apollo was not just a race with the Russian bear, but a powerful engine of progress relevant to other social goals. Webb eagerly complied, for he came increasingly to see NASA's real importance not in its space achievements but in its managerial effort and as a model for planned social change. At first, he wrote, the goals might have been set by the Soviet situation, but Apollo was not an end in itself. As progress was made, it became possible to enlarge and crystallize national goals. The

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Kennedy administration could be proud of having seen the danger of conservatism regarding new science and technology. "We must not repeat the failure to exploit the Wright Brothers . . . or Goddard." Of greater significance for the United States than proving its leadership in science and technology was "the motivation and drive of both the individuals and institutions of our society toward adaptation to modern requirements and self-improvement. Education will feel perhaps the greatest impact."²¹ In lectures around the country Webb went further. The space program was indispensable in all its aspects.

[T]he nations of the world, seeking a basis for their own futures, continually pass judgment on our ability as a nation to make decisions, to concentrate effort, to manage vast and complex technological programs in our own interest. It is not too much to say that in many ways the viability of representative government and of the free enterprise system in a period of revolutionary changes based on science and technology is being tested in space.²²

According to Webb, the space program required nothing less than the mobilization of the nation to a war footing in peacetime. Society (as he later wrote), had "reached a point where its progress and even its survival increasingly depend upon our ability to organize the complex and do the unusual. We cannot do these things except through large aggregations of resources and power." Whether we liked it or not, we were "in the midst of a crucial and *total* technological contest with the Soviet Union." What was being done in space could be done elsewhere in society, to engage "by all means available, in an endless search for new food for thought processes, new information and knowledge. . . ." Bigness was essential. Webb's "prototypes for tomorrow" included the Rockefeller oil empire, U.S. Steel, General Motors, the Panama Canal, TVA, and the Manhattan District. Large-scale endeavors were "adaptive, problem-solving, temporary systems of diverse specialists, linked together by coordinating executives in organic flux."²³

Space Age systems outgrew the resources of any institution and required the forging of a "university-industry-government complex"²⁴ for the waging of "war" on the technological frontier. This complex was not a necessary evil to Webb, but a positive boon. The product of large-scale endeavors was change—in the attitudes, interests, and concepts of reality of the people. "It follows that the larger the effort in science and technology, the larger those changes will be, and the more rapidly they will occur."²⁵ At NASA, Webb said, "they sought to minimize the disruptive effects of such a vast undertaking. For instance [I] rejected a proposal from G.E. that the lunar effort be turned over to it as prime contractor for the entire job, or from UCLA to become the 'concessionaire' for all scientific research involved in the space program."²⁶ Instead, NASA purposefully spread the wealth—90 percent of its funds flowed

into the private sector—and even pioneered the noncompetitive contract in order to save time and foster specific skills in a number of firms throughout industry. Likewise NASA “refused to go along with the old concept that scientific merit was the only determinant of who got a grant,” and favored second-rate universities even if the scientific merit of their proposals was “less than that of Caltech or MIT.”²⁷ Thus, NASA pioneered reverse discrimination in order to foster expertise in more regions (and please more congressmen). Webb granted that “traditions were broken,” but found it good: “In the working partnership between universities, industry, and government . . . each of the three has retained its traditional values. . . . I believe that each has become stronger because of the partnership.”²⁸

How was the country “stronger”? Not only because of economic spin-offs from large-scale endeavors, but because the government/industry/university team force-fed such spin-offs into the mainstream of society. Free enterprise might do that on its own, he granted, “but can society afford to wait for or to rely solely upon the workings of such a slow and uncertain process?”²⁹ Then there was the way in which new technologies altered human values. Webb quoted Raymond Bauer, whose study of the space program suggested that its effects “may include changes in man’s conceptions of himself and of God. . . .”³⁰ The space program promised a new era of great advances in the way large-scale efforts were managed, the encouragement of multidisciplinary efforts, new techniques and tools for the conduct of research in the social as well as physical sciences, and the manner in which they were applied to the solution of age-old problems.³¹ The burden fell on the administrators, like himself, to help subordinates to see the “totality of the job,” the relationship of their particular jobs to that totality, “to bring all their inner resources to bear in effective ways to help get the big job, the total job, done.”³²

What began as an extraordinary governmental initiative to reassure the world that individualism, free enterprise, and limited government were still superior came in Webb’s mind to be a vehicle for “revolution from above.” NASA’s destiny was to serve as prototype for reallocation of national power for social and political goals. “The technological revolution that is now fully upon us,” he wrote,

involves all areas and disciplines. It is the most decisive event of our times, and keeping ahead of it is essential. No nation that aspires to greatness, or to use its power for good, can continue to rely on the methods of the past. Unless a nation purposefully and systematically stimulates and regulates its technological advances and builds the fruits of those advances into the sinews of its system, it will surely drop behind. . . . The great issue of this age is whether the U.S. can, within the framework of existing economic, social, and political institutions, organize its development and use of advanced technology as effectively for its goals as can the Soviet Union. . . .³³

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The preceding quotations date from 1968-69, at the end of Webb's time at NASA. But as early as 1961 he had written Glennan that the basic institutions of the United States would have to be rethought in light of technological revolution. In 1963 he believed that

every thread in the fabric of our economic, social, and political institutions is being tested as we move into space. Our economic and political relations with other nations are being reevaluated. Old concepts of defense and military tactics are being challenged and revised. Jealously guarded traditions in our educational institutions are being tested, altered, or even discarded. Our economic institutions—the corporate structure itself—are undergoing reexamination as society seeks to adjust itself to the inevitability of change.³⁴

Nor was Webb alone in his belief that space technology would stimulate changes far beyond its sphere. Dryden predicted "a great variety of new consumer foods and industrial processes that will raise our standard of living and return tremendous benefits to us in practically every profession and activity."³⁵ Congressman Miller could "not think of any other aspect of our space program that could better justify our space expenditures to the average taxpayer than industrial applications. . . . His return will be a wide variety of new or better products, at reasonable cost, which in turn will give rise to a greater consumer demand and economic stimulus."³⁶ But space did more than spin off products like sparks from a flywheel. According to Edward Welsh of the NASC: "The aerospace revolution is a rebellion only in the sense that the forces at work do rebel against the old, the worn out, and the wholly ineffective methods and techniques. This is not change for the sake of change. . . ." Space technology did not permit even the slightest chance of unanticipated error, thus paper and pencil could no longer be tolerated where computers could work better and faster. "Now the process of systems analysis and the great experiments which have led to amazing advances in managerial competence are solving problems we previously thought impossible of solution. We are indeed enjoying a technological revolution and must not let it slow down." Space, said Welsh, would pay a greater dividend than anything else in which the United States engaged.³⁷

That the technocratic faith outgrew both the White House lust for prestige and the self-promotion of the Big Operators is indicated by the explosion of overall federal support for science, education, and R & D—and by the almost complete lack of congressional inquiry, oversight, or opposition to it. By FY 1964 the total NASA budget passed \$5 billion, five times what it was three years before. Counting the DoD, the AEC, and other agencies, the government spent \$6.8 billion on space, or a little over one percent of GNP. Total R & D performed in the United States rose over the three years from \$14.3 to \$18.9 billion, and now

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comprised 3 percent of the GNP. But government R & D rose from \$9.3 to \$14.7 billion, 58 percent above 1961 and now one-eighth of all federal spending. The trend begun under Eisenhower had quickened to the point that by FY 1964 over 78 percent of all American R & D was purchased by the federal government.³⁸

The sharp rise in federal R & D was concentrated, of course, in defense and space. So unless the United States were to transmogrify into a high-tech military economy while civilian industries went begging, the beneficiaries of the boom must demonstrate that R & D for space and defense also energized the larger economy. One result was that no agency studied its social and economic impact more vigorously than Webb's NASA. Regional development, technology transfer, the impact of R & D on growth, information diffusion systems, total employment generated—all in all, \$35 million were spent in NASA's first decade on research in the social sciences.³⁹ The results, accumulating in the later 1960s, were generally disappointing and mostly predictable. Economists demonstrated again that economic growth depended on too many things for the correlation between R & D and growth to be quantifiable. But the connection was intuitive, and Kennedy's sense that this was something the United States had to do was shared by many, even if the rewards might come in unfathomable ways. Perhaps they were right—unfathomable effects can hardly be shown *not* to exist. But to maintain space budgets and sustain Webb's larger plans for social renewal through large-scale management, justifications of the space program drifted naturally into the material realm.

The role of the space program in stabilizing the delicate aerospace industry was obvious—McNamara saw this in 1961. But Webb expected the space program, as integrator of social forces, to contribute to overall growth and stability. The kernel of the NASA university program, for instance, was mobilization of professorial talent for a conscious assault on social problems. Eisenhower had feared that the NDEA would outlast its temporary status and lead to federal funding and regulation of education at every level. But Webb—and the growing educationist establishment—affirmed such involvement and believed that strewing "seed money" would cause "excellence" to sprout and spread like hardy ground cover across the length and breadth of the country.

One of the persistent criticisms of Apollo was that it sucked up the nation's technical talent to the detriment of research in other fields. So Webb determined to give back what NASA was taking up, "to do all I can to build up the university research, teaching, and graduate and postgraduate quality and quantity of education."⁴⁰ The PSAC Panel on Scientific and Technical Manpower had called for the United States to graduate 7,500 scientists and engineers by 1970. By 1963 NASA's Sustaining University Program alone projected support for 4,000 graduate students per year, an effort equal in size to that of the NDEA science

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program. Only one or two congressmen suggested that NASA had exceeded its authority by embarking on any such program.⁴¹ But the real aim of the university program was social uplift. Webb wrote Lee DuBridge, president of Caltech, that as much as \$35 billion over ten years would be channeled through NASA (much of it to the West Coast) to advance science and technology at the most rapid rate, and "to feed it back into our national economy and the fabric of our national life." He sought to interest DuBridge in NASA-derived technology for urban development, water resources, energy, communications, management, and life sciences. "I know that the lunar objective causes some problem to you," but it would prove its worth "even if we never make the lunar landing."⁴² In January 1962 Webb promised Killian that NASA would not take professors away from their campuses, but find ways to use their talents while they remained with students and labs. He met with governors, university presidents, regional business delegations, and set up an ad hoc group to hold "think sessions" every month to plot the future of NASA/university/industry projects.⁴³

Did all this not smack of elitist *dirigisme*, or at least a failure to recognize that technology alone could not provide the value judgments on which wise direction of national energies should be based? Bush made this point, as did a generation of skeptics informed by C. P. Snow's *The Two Cultures*. Webb acknowledged that value judgments were involved, "but the result is also dependent upon the skill of the social organization in resource allocation," that is, the means must be developed for social mobilization before the question of ends was more than academic. The important point was that

we are so operating as to spread all of the problems, scientific and technical, over a very large number of able minds in educational institutions, in the media of communications and throughout industry. We might develop a body of doctrine here which indicates that the role of our senior leadership group is to so do this as to keep a fluid enough situation so that new concepts have an opportunity to flourish . . . and to be incorporated in the operating doctrines of our democracy through our present democratic processes or some incremental improvement or evolution of them.⁴⁴

The tool's the thing, the "very large scientific and technical capability" now in place through NASA's "ongoing contacts with industry and universities."⁴⁵

Those contacts went beyond hiring faculty as consultants and funding graduate students. NASA also sponsored grants for the construction of space sciences laboratories. University administrators made pilgrimages to Washington to solicit "brick and mortar" money. By the mid-1960s, the Office of Space Science and Applications devoted \$30 million per year to university research, while the sustaining university program dispensed another \$25 million. Twenty-seven facilities grants were made,

while professors, administrators, and students learned to ride the wave. The likes of Governor Otto Kerner of Illinois, the president of McDonnell Aviation in St. Louis, and a delegation from Los Angeles petitioned for space institutes at the University of Illinois, Washington University, and UCLA. When Professor Samuel Silver of Berkeley arrived to request a space science lab, Webb stipulated that the new center take on two economists to study the feedback of science and technology into the community, to solve social problems. By mid-1962 Webb made it a condition of support that universities seek "ways and means to assist its service area or region in utilizing for its own progress the knowledge, processes, or specific applications arising from the space program."⁴⁶

Such melding of theory and practice, of pure and applied research, of natural and social science, while hardly the same as the Soviet practice, nevertheless clashed with the traditional autonomy and values of American universities. In truth, the latter may never have taken the "social impact" clause of their contracts seriously, regarding it as congressional eyewash (this was NASA Space Science Director Homer Newell's suspicion). But that in itself suggests how far the country had come from 1945 when Congress *rebelled* against plans for central coordination of research to attack social problems. Now Congress *expected* such plans to justify funding. Even Newell granted that to place scholarly expertise "on ready call to be applied on command to problems of someone else's choosing . . . would destroy the very independence that generated the unique expertise in the first place."⁴⁷ But when academics dragged their feet on NASA-university assaults on local problems, Webb considered them "outrageously callous and irresponsible." For Webb, the organized pursuit of social objectives fixed by government was the whole point of the program.⁴⁸

Still, universities fell into step behind the aerospace firms and politicians, eager to accept federal largesse in the name of Webb's Space Age America. Conceived and promoted by politicians and intellectuals, endorsed by corporations, universities, and bureaucracies that stood to gain, not only military but civilian technocracy spread across the land with no more philosophical debate than accompanied the triumph of nuclear deterrence. Perhaps the United States had to give up much that was dear to meet the Soviet challenge. But why did so many make such a virtue of that necessity?

In April 1961, in the midst of the cloistered debate on whether to go to the moon, Senator Kerr left the Space Committee for a trip back to the hustings. It was Annual Law Day at the University of Oklahoma, and the Sooner State's favorite son was the honored speaker. His fortune was in oil, but his roots and votes were in the soil of '89—which blew away in '36—which made every Oklahoma politician's first responsibility the garnering of federal water projects. Indeed, his favorite oratorical

subject, said Kerr, was Land, Wood, and Water. But since chairing the Space Committee, he amended his longstanding slogan to Land, Wood, Water, and Space. "I believe this new version accurately describes a new horizon of great promise for the pioneering spirit of Oklahoma." Sooners had never lost the spirit of youth, optimism, and fearlessness toward what lay over the horizon: "... the qualities needed to explore the universe are the selfsame qualities that inspired our fathers and grandfathers to rush into Oklahoma's space and settle on what was, to them, a new and strange land."⁴⁹

Kerr slid easily into the big thinking inspired by the biggest frontier of all. But whence came James Webb's bold, technocratic vision? He was no midwestern Progressive, or "socialist of the chair" at some Ivy League university, or even a member of his party's left wing. Rather, Webb was a rural populist, who grew to know the workings of government and industry and their potential for huge concentrations of power. His own motive was service, his ability surpassing, and like LBJ he did not fear power. He had come to believe, like Kerr, that the United States was shaped by its frontier. Since 1896, the historians said, the domestic frontier was closed. But space was a new frontier in which American society might recover the rugged virtue that had almost been forgotten in the twentieth-century's mundane exploitation of the tamed land. Of course, space as the new frontier was a cliché of Kennedy's Camelot. But it was more than that for the Big Operators of the Southwest, the land of the Hoover Dam, Alamogordo, the Red River Project, White Sands, and now the Manned Spacecraft Center. Even as New Frontier slogans grew stale in public, they grew rich in meaning and implication for Webb until the whole nation, all its regions and institutions, became as a single band of homesteaders riding on rockets like Conestoga wagons.

In the midst of the Johnson Senate hearings after Sputnik, columnist James Reston's intuition served him well: the U.S. and Soviet governments alike would now have to prepare to deal with Texas.⁵⁰ Later on even von Braun, who exhibited the passion many Germans share for the lore and land of the American West, had to remark that "space is bigger than Texas."⁵¹ And then, in 1964 Webb discovered the theories of his namesake, Walter Prescott Webb, on the shaping of the American character on the frontier. In early 1965 he wrote to Horace Busby, a presidential aide, that problems such as Vietnam and racial integration and the other priorities on the national agenda were all of a piece. "I have several times around the country," he continued,

pointed out the frontier thesis. . . . based on the "wild and unperturbable" forces of the frontier, which show no mercy and no compassion, must be harnessed and utilized by the pioneer and in the process have the feedback effect of generating in the pioneer those qualities which have made for the American

democratic system, the same kind of analogy may be considered in connection with our large-scale organized efforts such as those in space. Here an entire nation is developing technology which puts it, as an organized entity, very much in the same position as the pioneer was individually on the frontier.⁵²

A world democracy could theoretically emerge as nations plied the frontier together—except that the Communists seemed disinclined to cooperate. "Whether we can force them to cooperate by developing so much power that there simply isn't any alternative is the real question before us *at this time*."⁵³

Here was an updated American vision—a Space Age America that was the pioneer writ large. "This may well put us into a position where we are more in control of the destiny of the world than we have been since the early years after World War II." If the Communists sought to fill vacuums like Korea, Vietnam, the Congo, then

we ourselves might then have a strong obligation to use our technology to begin to fill these vacuums. . . . Now what this may mean is that the government-industry-university team we have developed under the NASA system, which has been so effective in marrying science and technology and mobilizing large resources for focusing on limited but important objectives, might become the pattern needed by this nation. . . .⁵⁴

Webb discovered (albeit erroneously) that the original "frontier thesis" of Frederick Jackson Turner was published in 1903, the year of the Wright brothers' flight. Through such theses Webb found a role for NASA as heroic as that of the pioneers of aviation. Turner proclaimed the closing of the frontier, the end of that confrontation with impersonal, impassive, wild, gigantic Nature. In 1951 Walter Prescott Webb expanded the notion: all western civilization faced a closed frontier, and as a result "we have a great pain in the heart, and we are always trying to get it back again." But after Sputnik, he wrote with delight to fellow Texan Lyndon Johnson that he had apparently settled on the closing of the frontier too soon.⁵⁵ James Webb took it to heart: the Space Age changed everything. But where Khrushchev and the Soviets expected the new age to fulfill their dream of a future utopia, Webb and the American technocrats expected it to restore a past apotheosis. Thus when Webb spoke of total technological competition, unlimited power, and mobilization of the entire nation, it was not with the foreboding of the Yankee Henry Adams, who had foreseen and dreaded the twentieth century, but with the boundless enthusiasm of a Boomer Sooner and the lofty certainty of a Texas Longhorn.

Conclusion

The Kennedy years saw the quiet withdrawal of American political resistance to technocracy. Its triumph was all the more fundamental for its being so subtle.

In the late 1950s the principle of limited government still clung to life. Federal "brick and mortar" money for schools, aid to students in science and foreign languages, R & D for national defense and even for prestige: these were still defined as special violations of the hoary boundary between federal power and local freedoms. At the same time, the notion that these were not violations of that boundary, or indeed that the boundary itself was a superstition, made steady progress among northern Democrats, liberal Republicans, academics, journalists, and even conservative Cold Warriors—all of whom favored a technological "race posture" against the USSR, domestic social problems, or both. In 1961, as JFK proclaimed, the torch passed to the younger generation. Far from holding back the varied constituencies of an enlarged federal role, Kennedy relied on their coalescence and then picked from them the advisers that led the United States into the 1960s. An alliance of military hawks and social activists, they endorsed federal action on the principle that the state could best foster new knowledge, power, and economic growth through planned management. Once in power, the constituencies began to quarrel, each favoring vigorous federal action in some areas and not others. But by 1964 the fundamental question—whether a society shaped by state planning and spending was consonant with American freedoms—got lost. Liberals wanted less hardware and more social engineering; mainstream Democrats insisted that the United States could afford both; even Goldwaterites granted government's redoubled responsibility for Space Age technology, if only for national defense. The debate revolved around the purposes to which the technocratic tool ought to be put, not the political or moral costs of brandishing the tool in the first place.

The roots of American technocracy were as varied and deeply buried as its visible branches were varied and widespread. The doctrine of efficiency dated from the Progressive Era, the protest against the insecurities of the free market from the depression, the temptation to forge government/industry/university teams for federally directed progress from World War II, the demand for equality for the underprivileged

from the postwar years. Perhaps these tendencies would have nudged the United States into technocracy in any case, according to the American tradition of softening social or ideological discord through expansion and growth—the frontier policy. But the challenge of Sputnik, because it symbolized Soviet parity in strategic weapons, and Khrushchevian peaceful coexistence (read “nonviolent competition”), because it pushed Cold War rivalry into every arena of national achievement, surely helped to trigger the rapid political change of 1957 to 1964. Propaganda to the effect that racism did not exist in the USSR, that Soviet women were liberated, that its economy would soon surpass the American, that the social product was equally distributed among Soviet citizens, might all be cynical lies. But Americans had difficulty damning the USSR while headlines proclaimed riots in Birmingham and the rhetoric of every disaffected social group. Above all, the Soviet claim on the future through scientific and technical supremacy could hardly be gainsaid while cosmonauts beat the Yanks to every spectacular.

Throughout the century, Soviet leaders declared that the Western countries might be ahead for the moment but were run by and for monopolists who lived off imperialism foisted on others by their high technology. The USSR, on the other hand, was a people's state . . . and in any case would soon catch up. Now the United States was saying that the Soviet Union might be ahead for the moment but was run as a slave state in which resources could be mobilized to serve militaristic ambitions. The United States, on the other hand, was a free country . . . and in any case would soon catch up. In so saying and doing, American leaders bought the Communist line that technology was both a symbol of social superiority and the main agency for making one's own preferred image real. The Midas touch of government-directed R & D, education, and social engineering could turn base society into a Great Society, eliminate want, inequality, injustice, and even, in LBJ's vision, dispel hatred.

Through all this, U.S. space policy changed little under Kennedy. Government assumed a larger role in comsat deployment, but positions on space law and cooperation—and the contradictions of those positions—remained. The military program fell under tighter wraps and civilian control. The American agenda at the UN triumphed in the Principles of 1963: national sovereignty and nuclear bombs banned in space, but no inspection of space launches to ensure that they were for “peaceful purposes,” no demilitarization, no UN space agency, and no definition of where outer space began. American (and Soviet) policy makers judged all this prudent and were probably wise to preserve a loose regime in which spaceflight could advance most rapidly and serve the national interest. Similarly, U.S. agencies continued a deliberate approach to cooperation that stemmed from the inherent difficulties of international R & D and the fact that competition was the motive force for space

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spending in the first place. The preferred posture was cooperation in science, competition in engineering. This did little to please Europeans, who were coming to believe Superpower rhetoric about space R & D as a forcing house of technology and growth, and resented U.S. dominance. The Kremlin never showed serious interest in cooperation. Kennedy's call for a joint moon program reverberated, but its sounding boards were the UN and Congress, not the Kremlin.

Thus the space technological revolution did not change world politics or usher in an age of global integration or even an abatement of the Cold War. The international system absorbed space just as it absorbed the atom. Nor did the Space Age oblige traditional elites to surrender policy to a technical elite. Ike's fears of a military-industrial, scientific-technological complex were not confirmed in the way he expected. The danger was never a *technocracie* of technicians, but a technocracy of politicians, arrogating to government the right to fix a national agenda and order fabrication of techniques, both hardware and management, for its fulfillment. Even NASA ceased to be an agency run primarily by scientists and became a juggernaut of the politicians and engineers. The military chiefs lost what remained of their autonomy in military strategy, R & D, and procurement to the civilian managers. University scientists, businessmen and technicians in the corporations, these, too, were integrated in varying degrees into a national complex for the promotion of new techniques, paid for, directed by, and dependent on the whim of the federal government.

Kennedy knew little about space, but he bowed to politics, as if intuitively, and pointed to the moon. Within two years he became a believer. "But why, some say, the moon? Why choose this as our goal?" he asked an audience at Rice University in September 1962. "Why climb the highest mountain? Why, 35 years ago, fly the Atlantic? Why does Rice play Texas?"¹ To aspire, to strive, to slay giants. The challenge of the frontier now lay in technology and, physically, in outer space. If the effort required the nation to become as one man, like Lindbergh fixed to his machine like a knight errant, or as a collective, like a well-oiled, inspired football team pitted against a bigger rival—then so be it. At least we, in the United States, would do it voluntarily, and not by coercion.

James Webb, too, became a believer. Once he moved enough rocks from his imposing pile to see that a moonflight might just be achieved on schedule, he marveled at the power of organization it revealed. LBJ, in Webb's judgment, was not a good administrator. Nor were Kennedy or Eisenhower.² But Webb himself was, and like McNamara he believed that the real frontier was a management frontier. The space program promised many ineffable but one certain benefit—perfection of the management of large systems. Apollo was not an end, but a means that brought order out of the chaos of institutions and that could be applied

to a host of social projects. Webb himself may never have chanted, "If we can send a man to the moon, why can't we . . . ?" But if others did, calling for Apollo-type assaults on everything from medical care to synthetic fuels, it was because the James Webbs had, by their talent and energy, made command innovation look easy—and "American."

Soon after Johnson's landslide, Webb meditated on technocracy American-style. "Progress in science, in technology, and in efficient organization of large-scale effort is of itself not sufficient; any great society must as well believe in the values of intellectual vigor and the creativity of the individual." Good Americanism, but how was this to be preserved? "The work of individuals," Webb wrote the President,

must be organized within a framework. . . . The leisure of the past is gone; where it once took hundreds of years to fully exploit a new technology, we are now capable of taking enormous strides in a very short time. . . . The earth orbiting satellite has become the first tool of the human race that is not limited by the fuel it can carry, the boundaries of other nations, or by the earth's atmosphere or its oceans. The fact that it can work for any nation over which it passes is a symbol of a great society of the world, and, perhaps more than any tool before, a portent of a more universal society.³

Space spending would not detract from other things in the building of a great society.

In fact, greater support of the space program will undoubtedly yield greater support for these other needs by demonstrating what can be done when we find new and more effective ways to involve large numbers of imaginative, gifted, and innovative people in an area that has almost explosive potential. . . . In reality, the space program lies in your first area of building the great society.⁴

Johnson himself was far less moved by the adventure and wonder that moved Kennedy—the wonder that still, in the National Air and Space Museum, brings tears to the eyes of more visitors than the Declaration of Independence or Lincoln Memorial. To LBJ the space program was a model of the role government should play in society, and the role technology should play in government, an expression as well of new and apparently limitless power, as if handed down by some Promethean party boss in the form of command technology and federal management. For the War on Poverty, the Great Society, as much as Apollo or Vietnam, were Cold War phenomena, but they were not only that, far from it. They were born of a moral vision in which men of power and charity sought to use their gifts for the less gifted. It was possible to eradicate poverty, crime, ignorance, whip the Communists, and develop the Third World, or so thought LBJ. The power existed and needed only to be grasped. Indeed, it must be grasped—by good men—lest the other side do so first. To Eisenhower, the essence of courage was

Conclusion

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to resist the temptation to use dangerous tools; to Johnson, the essence of courage was to dare to take them up in a good cause. Whether in decaying cities, outer space, or Third World jungles, American technology would overwhelm the enemies of dignity.

For several years after the 1964 election, technocratic enthusiasm spread with the appeal of a perpetual motion machine. Adlai Stevenson called it "our almost miraculous capacity to use existing technology to create new technology" and declared that "Science and technology are making the problems of today irrelevant in the long run, because our economy can grow to meet each new charge placed upon it. . . . This is the basic miracle of modern technology. . . . It is in a real sense a magic wand that gives us what we desire. Don't let us miss the miracle by underestimating this fabulous new tool."⁵

Miraculous, magical, fabulous—it was not the scientists and engineers who were euphoric at the prospect of thaumaturgy, but the politicians and their allies in academics, business, and journalism. The euphoria fed a sixfold increase in federal R & D over a single decade, then an expansion of social engineering that proved as irreversible as a technological revolution. Civilian space spending peaked in FY 1965 ("Some people's feet got tired," recalled LBJ). But space spending was no longer the measure of the technocratic trend, for the mentality and methods of the space program had taken hold elsewhere: "We realized, after *Sputnik I* and *II*, that we had been challenged and challenged successfully. . . . Our people are slow to start but hard to stop." Those with tired feet "raised the question, 'Well, if we can go to the moon, why don't we take that money and do some of the things that need to be done here?'" Until Sputnik, said Johnson,

the Federal Government hadn't passed any education bill. We didn't have any Federal aid for education. . . . So we started passing education bills, we made a national effort in elementary education, a national effort in higher education, where two million students were brought into our colleges. And they said, "Well, if you do that for space and send a man to the moon, why can't we do something for grandma with medicare?" And so we passed the Medicare Act, and we passed forty other measures. . . .

And I think that's the great significance that the space program has had. I think it was the beginning of the revolution of the '60s.⁶

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58. K. Sergeyev in *Pravda*, 1 Jan. 1964 and 1 Jan. 1965, cited by Daniloff, *Kremlin and the Cosmos*, pp. 116-17. Even poor Korolev's name was taboo. He published New Year's articles annually in *Pravda* under the pseudonym Professor K. Sergeyev. The same formula was used for his "deputy," Mikhail Kuzmich Yangel, who became in print "Vasily Mikhailovich."

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61: HST Library,

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well as the complaints of Medaris and Gavin cited in chapters 8 and 9 herein.

16. Memo, Thomas D. White (USAF Chief of Staff) to Generals Landon, Wilson, LeMay, Schriever, Dr. Perkins, and the Undersecretary of the USAF, 14 Apr. 1960, NASA HO:

I am convinced that one of the major long range elements of the Air Force future lies in space. It is also obvious that NASA will play a large part in the national effort in this direction and, moreover, inevitably will be closely associated, if not eventually combined with the military. It is perfectly clear to me that particularly in these formative years the Air Force must, for its own good as well as for the national interest, cooperate to the maximum extent with NASA.

17. Lt. Col. Kenneth F. Gantz, ed., *The U.S. Air Force Report on the Ballistic Missile: Its Technology, Logistics, and Strategy*, preface by General Thomas D. White, intro. by Major General Bernard Schriever (New York, 1958); Kenneth F. Gantz, *Man in Space: The United States Air Force Program for Developing the Spacecraft Crew* (New York, 1959); Editors of Air Force Magazine, *Space Weapons: A Handbook of Military Astronautics* (New York, 1959). *Man in Space* even argued for a moon program under military auspices. If the project proved to be impossible, "we shall know that no one else can succeed"; if it proved possible, "This outpost, under our control, would be the best possible guarantee that all of space will indeed be preserved for the peaceful purposes of man" (p. 253).

18. Letter, Schriever to Gardner, 11 Oct. 1960, NASA HO. Schriever compared the proposed study to the von Neumann committee of 1954. Gardner's associates included William O. Baker (vice president for research of Bell Labs), Harold Brown (of the University of California's Lawrence Radiation Lab), William C. Foster (vice president of Olin Mathieson Chemical Corp.), and eleven other luminaries from science, business, and the USAF. They were advised by a special study group of scientists and engineers convened at Los Alamos, chaired by Dr. Theodore B. Taylor (Senior Research Adviser of General Dynamics' Atomic Division).

19. Office of the Secretary of the USAF, Memo to commanders and contractors, "Air Force Competency in Space Operations," 1 Dec. 1960, in U.S. Congress, House Committee on Science and Astronautics, *Defense Space Interests*, 87th Cong. 1st Sess., 1961, pp. 93-96.

20. "Report of the Air Force Space Study Committee" ("Gardner Report"), HQ, ARDC, 20 Mar. 1961, NASA HO. Quotes from pp. 2, 3, 4, 54. Italics in original. The lunar base proposal drew on a previous USAF project calling for the integration of NASA and military requirements into one national lunar program, with establishment of a lunar base to be considered a military expedition (Directorate of Space Planning and Analysis, USAF, Ballistic Missile Division [BMD] "Military Lunar Base Program or S.R. 183 Lunar Observatory Study," Apr. 1960, NASA HO).

21. Memo, Overton Brooks, "Attitude of Committee on Science and Astronautics Relative to the National Space Program," 13 Feb. 1961, NASA HO.

22. Letter, Brooks to the President, 9 Mar. 1961, NASA HO.

23. U.S. Congress, House Committee on Science and Astronautics, *Defense Space Interests*, 87th Cong., 1st Sess. (March 1961), pp. 35 (Gilpatrick quote), 92-93 (General White testimony). See also the committee's conclusions in U.S. Congress, Committee on Science and Astronautics, *Military Astronautics (Preliminary Report)*, 87th Cong. 1st Sess. (May 1961), pp. 35-37.

24. Letter, Kennedy to Brooks, 23 Mar. 1961, NASA HO.

25. Schriever Speech before the Allegheny Conference on Community Development, 21 Nov. 1960, reprinted in *Congressional Record—Appendix*, 1961, pp. A93-94. On NASA-DoD relations generally, see also Richard H. Campbell, "The Impact of NASA on the Air Force Role in Space," Air Command and Staff College thesis, Maxwell AFB, Ala., 1965, and Stephen Ira Grossbard, "The Civilian Space Program: A Case Study in Civil-Military Relations," Ph.D. diss., University of Michigan, 1968.

26. Memo, Hyatt, "Some Notes on the Relationship of NASA-DoD Space Programs," 19 Apr. 1961, NASA HO.

27. NAS, Space Science Board, "Man's Role in the National Space Program," p. 1, undated (resulting from meetings of 10-11 Feb. 1961) and forwarded in Letter, Berkner to Waterman (NSF), 31 Mar. 1961; NASA HO. Italics in original. The document was re-released, with reference to the February meetings, on 7 Aug. 1961, and may be found in U.S. Senate, Committee on Aeronautical and Space Sciences, *National Goals for the Post-Apollo Period*, 89th Cong., 1st Sess. (Aug. 1965), pp. 242-43. On the Space Science Board's role in 1959-61, see also the brief accounts by Charles M. Atkins, "NASA and the Space

Science Board of the National Academy of Sciences" NASA, HHN-62, Office of Policy Analysis, Sept. 1966, and Logsdon, *Decision To Go to the Moon*, pp. 88-89.

28. NAS, Space Science Board, "Man's Role in the National Space Program," p. 3.

29. Homer E. Newell, Memo for Files, "Notes on the SSB Meeting," 10-11 Feb. 1961, NASA HO. Those who held manned spaceflight to be misdirected included Joshua Lederberg, Donald Hornig, and Willis Shapley (from BoB); Berkner on "negativism" (p. 3) and "pull together a statement" (p. 4). On Webb hearing of the SSB action a month early (in late February), see Logsdon, *Decision To Go to the Moon*, p. 89.

30. Memo, BoB, Military Division, to Bell (Director, BoB), "Meeting with Mr. James E. Webb," 15 Feb. 1961, NASA HO.

31. Jay Holmes, *America on the Moon: The Enterprise of the Sixties* (Philadelphia, 1962), p. 194; Logsdon, *Decision To Go to the Moon*, pp. 89-90.

32. Webb Speech to the American Astronautical Society, 17 Mar. 1961, cited by Logsdon, *Decision To Go to the Moon*, p. 90.

33. Memo, Bell for the President, "National Aeronautics and Space Administration budget problem," undated (describes 22 Mar. 1961 meeting), NASA HO.

34. Quoted both by Holmes, *America on the Moon*, p. 194, and Logsdon, *Decision To Go to the Moon*, p. 91.

35. Robert Jastrow (NASA-Goddard Space Flight Center), "Evaluation of USSR vs. US Output in Space Physics," 27 Feb. 1961, NASA HO.

36. "Administrator's Presentation to the President," 21 Mar. 1961, NASA HO.

37. Interviews with David Bell and Edward Welch by John Logsdon, *Decision To Go to the Moon*, p. 100.

38. *Washington Post*, 13 Apr. 1961; *New York Times*, 16-17 Apr. 1961.

39. U.S. Congress, House Committee on Science and Astronautics, *Hearing on H.R. 6169—A Bill to Amend the National Aeronautics and Space Act of 1958*, 87th Cong., 1st Sess. (April 1961), Fulton quote from p. 5.

40. U.S. Congress, House Committee on Science and Astronautics, *Discussion of Soviet Man-in-Space Shot*, 87th Cong., 1st Sess. (April 1961), Fulton quote from p. 7, Anfuso quote from p. 13. On "tired of coming in second-best," see Ken Hechler, *The Endless Space Frontier. A History of the House Committee on Science and Astronautics, 1959-78* (Washington, D.C., 1980), p. 82.

41. Hechler, *The Endless Space Frontier*, p. 83.

42. The President's News Conference, 12 Apr. 1961, *PP of P, John F. Kennedy 1961*, pp. 262-63.

43. NASA Press Conference on Russian Space Shot, 12 Apr. 1961, cited by Logsdon, *Decision to Go to the Moon*, p. 105.

44. Hugh Sidey, *John F. Kennedy, President* (New York, 1964), pp. 121-23.

45. Interview with James Webb by Logsdon, *Decision To Go to the Moon*, p. 109. The Bay of Pigs itself was apparently not a factor in the lunar deliberations, at least in the documentary record. Ed Welsh insists that it was never brought up by anyone in the Kennedy counsels (Edward C. Welsh Oral History, JFK Library, p. 20).

46. Letter, Webb to Glennan, 12 Apr. 1961, NASA HO.

47. Memo, Kennedy for the Vice President, 20 Apr. 1961, NASA HO.

48. The President's News Conference, 21 Apr. 1961, *PP of P, John F. Kennedy 1961*, pp. 309-10.

49. Memo, NASA for the Vice President, 22 Apr. 1961, NASA HO.

50. Logsdon, *Decision To Go to the Moon*, pp. 115-16. Cook was Executive Vice President of American Electric Power Corporation.

51. Memo, Johnson for the President, "Evaluation of the Space Program," 28 Apr. 1961, NASA HO.

52. Richard Hirsch and Joseph John Trento, *The National Aeronautics and Space Administration* (New York, 1973), p. 108.

53. Letter, Webb to LBJ, 26 Aug. 1966, LBJ Library, White House Central Files, Outer Space; "close to demanding," from Logsdon, *Decision To Go to the Moon*, p. 120.

54. Letter, Webb to LBJ, 4 May 1961, NASA HO.

55. Memo, Colonel Burris to the Vice President, "Views of Secretary McNamara on Acceleration of Booster Capabilities and Manned Exploration in Space," 23 Mar. 1961, NASA HO.

56. Bright, *The Jet Makers*, pp. 169-73; William L. Baldwin, *The Structure of the Defense*

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Market 1955-1964 (Durham, N.C., 1967); Tae Saeng Shin, "A Financial Analysis of the Airframe-Turned-Aerospace Industry," Ph.D. diss., Univ. of Illinois, 1969 (on business and financial risk, p. 146).

57. Charles D. Bright, *The Jet Makers* (Lawrence, Kans., 1978), pp. 70-73. The breakdown of aerospace profits and losses from 1958 to 1960 was found in *Aviation Week and Space Technology*, 12 June 1961, p. 115. Only three major manufacturing sectors in the economy had lower composite profit margins than aerospace.

58. Major-General Ostrander (USAF Director, Launch Vehicle Programs) had already gotten word of a planned moon program and recommended that it be given to NASA on the condition that responsibility for all other launcher programs, meteorological and communication satellites, and orbital operations be given to the USAF. Memo, Ostrander for Robert Seamans, "Reflections on the Present American Posture in Space," 21 Apr. 1961, HST Library, Webb Papers, Box 103.

59. Willis Shapley, National Air and Space Museum Seminar, 11 Feb. 1982. Shapley reports that no one was more stunned than young Berkeley physicist Harold Brown, summoned to discuss the moon his first day on the job at the Pentagon! Uncomfortable in his California sportcoat amid all the gray suits, he said nothing. Brown became Secretary of Defense under Jimmy Carter. Overton Brooks was also invited to the meeting but did not attend. He expressed his views in a long memo, "Recommendations re the National Space Program," 4 May 1961, attached to Letter, Brooks to LBJ, 5 May 1961, NASA HO.

60. Logsdon, *Decision To Go to the Moon*, p. 124.

61. Willis Shapley, NASM Seminar, 11 Feb. 1982.

62. "Recommendations for Our National Space Program: Changes, Policies, Goals" (the Webb-McNamara Report), May 1961, JFK Library, Presidential Office Files. Quotes from pp. 10, 11, 18, 25.

63. Memo, Shapley to Bell, "Staff Review of Proposed Increase in Space Programs," 18 May 1961, NASA HO.

64. McGeorge Bundy Interview by Logsdon, *Decision To Go to the Moon*, p. 126; Jerome Wiesner Interview by E. M. Emme and Alex Roland, NASA HO.

65. Letter, Webb to LBJ, 23 May 1961, NASA HO.

66. See, for instance, Howard Margolis, "Technological and Political Models of the Decision To Go to the Moon," Institute for Defense Analyses—International and Social Studies Division (Sept. 1972), who argued (once the jargon is penetrated) that *both* of his models of decision making account for Apollo. "The key question was how a decision of this magnitude could have been made without a serious technical review" (p. 174).

Chapter 16

1. On the Kennedy-McNamara relationship, see William W. Kaufmann, *The McNamara Strategy* (New York, 1964), pp. 3, 300; on the whiz kids and civil management of defense, pp. 31-33; on the "unfinished business" inherited from Eisenhower, see Alain Enthoven and K. Wayne Smith, *How Much Is Enough? Shaping the Defense Program, 1961-1969* (New York, 1971), pp. 1-30.

2. See Richard Fryklund, *100 Million Lives: Maximum Survival in a Nuclear War* (New York, 1962), pp. 1-16; Peter Pringle and William Arkin, *The SIOP. The Secret U.S. Plan for Nuclear War* (New York, 1983), pp. 101-8 (Eisenhower reference, pp. 107-8); David Alan Rosenberg, "The Origins of Overkill: Nuclear Weapons and American Strategy, 1945-1960," *International Security* 7 (Spring 1983).

3. Desmond Ball, *Politics and Force Levels: The Strategic Missile Program of the Kennedy Administration* (Berkeley, 1980), pp. 34-38. This is the latest and best researched history of American strategy and missile deployment in the 1960s. The following discussion depends on it heavily. But see also Elie Abel, *The Missile Crisis* (New York, 1966); Richard J. Barnett, *The Economy of Death* (New York, 1969), a leftist attack on the military-industrial complex; C. W. Borklund, *The Department of Defense* (New York, 1968); Edgar M. Bottome, *The Missile Gap: A Study of the Formulation of Military and Political Policy* (Cranbury, N.J., 1971); Enthoven and Smith, *How Much Is Enough?* by the chiefs of McNamara's Systems Analysis Office; Roger Hilsman, *To Move a Nation* (New York, 1964), and *The Politics of Policy-Making in Defense and Foreign Affairs* (New York, 1971); Samuel P. Huntington, *The Common*

Chapter 18

1. See, for example, Bruce Mazlish, ed., *The Railroad and the Space Program; An Exploration in Historical Analogy* (Cambridge, Mass., 1965); John Logsdon, *The Decision To Go to the Moon: Project Apollo and the National Interest* (Cambridge, Mass., 1970); Vernon Van Dyke, *Pride and Power: The Rationale of the Space Program* (Urbana, Ill., 1964); Amitai Etzioni, *The Moon-Doggle, Domestic and International Implications of the Space Race* (Garden City, N.Y., 1964); and Norman Mailer, *Of a Fire on the Moon* (New York, 1969) for varying interpretations of why Apollo was somehow very "American." It is interesting that, through the years, almost no commentators, whether they were for or against it, thought Apollo to be a rather "un-American" novelty in our history.

2. Jane Van Nimmen and Leonard C. Bruno with Robert L. Rosholt, *NASA Historical Data Book, 1958-1968*, 2 vols., vol. 1: *NASA Resources* (Washington, D.C., 1976).

3. "Reappraising NASA's Organizational Structure to Achieve the Objectives of an Accelerated Program," 12 June 1961, NASA HO. See also Robert L. Rosholt, *An Administrative History of NASA, 1958-1963*, NASA SP-4101 (Washington, D.C., 1966), pp. 209-11, and Arnold S. Levine, *Managing NASA in the Apollo Era*, NASA SP-4102 (Washington, D.C., 1982), pp. 1-4, 34-43.

4. Memo, Webb for the Vice President, 23 May 1961, pp. 1-2, NASA HO.

5. *Ibid.*, p. 2.

6. Ken Hechler, *The Endless Space Frontier. A History of the Committee on Science and Astronautics, 1959-1978* (Washington, D.C., 1980), p. 178. On the location of NASA facilities, see Thomas P. Murphy, *Science, Geopolitics, and Federal Spending* (Lexington, Mass., 1971), pp. 197-223. The biggest congressional tiff was touched off by the abrupt NASA decision to locate a new Electronics Research Center on Route 128 outside of Boston, the preferred choice of a number of prominent Democrats (Murphy, chap. 8).

7. Hugo Young, Bryan Silcock, and Peter Dunn, "Why We Went to the Moon," *Washington Monthly* 2 (Apr. 1970): 48. Their allegations are corroborated in the memoirs of Bobby Baker himself (with Larry L. King), *Wheeling and Dealing* (New York, 1978), pp. 103-04, 169-70. On the selection of Houston for the Manned Spaceflight Center, see the comparison of eighteen cities in Mary Holman, *The Political Economy of the Space Program* (Palo Alto, Calif., 1974), p. 50, and Webb's Memo to Dryden and Seamans, 31 Jan. 1963, citing the reasons for their judgment in favor of North American, HST Library, Webb Papers, Box 33. Houston tied for first on objective criteria with Los Angeles, San Diego, San Francisco, and New Orleans. On politics in the selection of NASA sites and award of contracts, see also Young, Silcock, and Dunn, "Why We Went to the Moon," and Hechler, *The Endless Space Frontier*, p. 185. The North American Aviation contract became a favorite of scandalmongers in 1967 when astronauts Grissom, White, and Chafee were incinerated in the test Apollo command module. See U.S. Congress, Committee on Science and Astronautics, Subcommittee on NASA Oversight, *Investigations into the Apollo 204 Accident*, 90th Cong., 1st Sess., 1967.

8. The public image of NASA and the astronauts is the commonest theme of popular writing about the space program. In most cases the authors aggressively "humanize" the astronauts and engineers (not a difficult task, their being human), only to reveal qualities more endearing than those concocted by the public relations men themselves. See especially Tom Wolfe, *The Right Stuff* (New York, 1979), and James Michener, *Space* (New York, 1982).

9. On the 1961 managerial reforms, see Rosholt, *An Administrative History of NASA*, pp. 209-39; Levine, *Managing NASA*, pp. 34-46. These 1961 reforms were not wholly successful. The functional lines of authority meant that one center might be involved in projects divided among five different program offices. The burden fell on Seamans, who could not untangle all the strings by himself. In 1963 Webb decentralized management, promoting project managers to autonomy as associate administrators for their own offices.

10. Levine, *Managing NASA*, pp. 68-93; quotation from p. 53. On PERT, see pp. 156-57. But Harvey M. Sapolsky, *The Polaris System Development. Bureaucratic and Programmatic Success in Government* (Cambridge, Mass., 1972), considers PERT overrated even in its debut with the navy (pp. 94-130).

1. *An Exploration on To Go to the Moon* Van Dyke, itai Etzioni, *The den City, N.Y.*, 19) for varying ng that, through ought Apollo to

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y of NASA, pp. olly successful. ed in projects who could not ent, promoting es. ee pp. 156-57. l *Programmatic* ed even in its

11. On the heated eighteen-month debate on the lunar mode, see John Logsdon, "Selecting the Way to the Moon: The Choice of the Lunar Orbital Rendezvous Mode," *Aerospace Historian* 18 (June 1971): 66-70; Holman, *Political Economy of the Space Program*, pp. 98-102; Roger E. Bilstein, *Stages to Saturn: A Technological History of the Apollo/Saturn Launch Vehicles*, NASA SP-4206 (Washington, D.C., 1980), pp. 60-68; Courtney Brooks, James Grimwood, and Loyd S. Swenson, Jr., *Chariots for Apollo: A History of Manned Lunar Spacecraft*, NASA SP-4205 (Washington, D.C., 1979), pp. 61-86. The motives of George Low and Wernher von Braun in finally opting for LOR are still obscure. Each had held tenaciously to the other modes. Von Braun himself thought it important to develop a capacity for orbital rendezvous, important as it was to the future of space development. On the other hand, von Braun wanted to continue building ever larger rockets. If he had known that choosing LOR meant the end not only of Nova but even of the *Saturn 5* after 1972, perhaps he would have continued to fight. There is speculation that a deal might have been struck, for von Braun had remained wedded to EOR until the climactic Huntsville meeting itself (7 June 1962), when he suddenly rose to endorse LOR, thereby undercutting his own people: "You could hear a pin drop." It is also possible, however, that von Braun simply wearied of the fight: "We are already losing time in our overall program," he said, "as a result of lacking a mode decision."

12. The heated public exchange in September 1962 has been oft described. Von Braun was hosting the presidential party around Marshall Spaceflight Center and explaining the LOR sequence when Kennedy interrupted, "I understand Dr. Wiesner doesn't agree with this. Where is Jerry?" Wiesner approached and was obliged to register his objection. But the debate occurred out of earshot of reporters, and Kennedy soon called it to a halt.

13. Webb acknowledged that ancillary benefits of potential military value to be derived from EOR versus LOR were an issue of great importance that required detailed discussion with the DoD (Memo, Webb to Brainerd Holmes, 8 June 1962, NASA HO).

14. Holman, *Political Economy of the Space Program*, p. 99; Harold L. Nieburg, *In the Name of Science* (Chicago, 1966), p. 29; U.S. House, Committee on Science and Astronautics, *NASA Lunar Orbit Rendezvous Decision*, 87th Cong., 2nd Sess., July 1962.

15. Loyd S. Swenson, Jr., James M. Grimwood, and Charles C. Alexander, *This New Ocean. A History of Project Mercury*, NASA SP-4201 (Washington, D.C., 1966), p. 477.

16. *Ibid.*, pp. 492 ("trust in God"), 420 ("flying camera"), 503 ("dust on highway").

17. Van Nimmen, Bruno, with Rosholt, *NASA Historical Data Book*, vol. 1, p. 134. Total space spending is recounted annually in the rear of the *Aeronautics and Space Report of the President*, published annually by the NASA HO.

18. Letter, Webb to the President, 29 Oct. 1962, NASA HO.

19. Letter, Webb to the President, 30 Nov. 1962, NASA HO.

20. Memo, Kennedy for the Vice President, 9 Apr. 1963, NASA HO.

21. Letter, Webb to the Vice President, 3 May 1963, HST Library, Webb Papers, Box 34.

22. Address by James Webb, Institute of Foreign Affairs, 24 Jan. 1963, p. 2, NASA HO.

23. James Webb, *Space Age Management: The Large Scale Approach* (New York, 1969), pp. 15, 17, 22, 29, 33-37.

24. Address by Webb, Milwaukee Press Club Gridiron Dinner, 17 Apr. 1963, in NASA, *The American Space Program. Its Meaning and Purpose*, HHN-47 (Dec. 1964), p. 25.

25. Webb, *Space Age Management*, p. 113.

26. *Ibid.*; see also Interview with Webb by Dr. Harvey et al., 16 Mar. 1968, pp. 16-17, NASA HO.

27. Interview with Webb by Dr. Harvey et al., 16 Mar. 1968, p. 16, NASA HO.

28. Webb, *Space Age Management*, p. 117.

29. *Ibid.*

30. Raymond Bauer, *Second Order Consequences: A Methodological Essay on the Impact of Technology* (Cambridge, Mass., 1969), pp. 3-4. Webb cites Bauer in *Space Age Management*, p. 121.

31. Webb, *Space Age Management*, pp. 123-24.

32. *Ibid.*, p. 151.

33. *Ibid.*, pp. 16-17.

34. Webb in "Conference on Space, Science, and Urban Life," Dunsmuir House, Oakland, Calif., 30 Mar. 1963, in NASA, *The American Space Program. Its Meaning and Purpose*, p. 33.

35. U.S. Senate, Committee on Appropriations, *Independent Offices Appropriations*, 1962, 87th Cong., 1st Sess., 1961, p. 649.
36. *Congressional Record*, 17 July 1963, p. A4508.
37. Welsh Speech, "Benefitting from Aerospace Revolution," Institute of Electrical and Electronics Engineers, Schenectady, N.Y., 17 May 1969, pp. 1-2. See also Welsh Speech, "Space and the National Economy," Conference on Space Science and Space Law, Norman, Okla., 20 June 1963.
38. National Science Foundation, *Federal Funds for R & D*, NSF 79-310, vol. 27, appendices.
39. Holman, *Political Economy of the Space Program*, esp. chaps. 5 and 9.
40. Letter, Webb to Vannevar Bush, 15 May 1961, HST Library, Webb Papers, Box 67.
41. Van Dyke, *Pride and Power. The Rationale of the Space Program*, pp. 224-30; Homer E. Newell, *Beyond the Atmosphere: Early Years of Space Science*, NASA SP-4211 (Washington, D.C., 1980), pp. 225-28.
42. Letter, Webb to DuBridge, 29 Aug. 1961, NASA HO.
43. Letter, Webb to Killian, 10 Jan. 1962, Box 31; Memo, Webb for Dryden, Seamans, and Simpson, 20 May 1964, Box 164: both HST Library, Webb Papers. Note: Most of Webb's outgoing correspondence from his tenure as NASA administrator can also be found in the Chron File at the NASA HO.
44. Memo, Webb for Dryden, 19 June 1964, HST Library, Webb Papers, Box 165.
45. Letter, Webb to Mose L. Harvey (Director, Center for Advanced International Studies, Univ. of Miami, Fla.), 29 June 1964, HST Library, Webb Papers, Box 36.
46. Newell, *Beyond the Atmosphere*, pp. 232-34; quotation from p. 233.
47. *Ibid.*
48. *Ibid.*; see also Webb, "Education in the Space Age," *Educational Record* (Winter 1964): 33-40.
49. Kerr Speech, "The Challenge of Outer Space to the Legal Profession," Norman, Okla., 27 Apr. 1961, NA, Papers of the Senate Space Committee, Box 102.
50. James Reston, "Senator Johnson's Move," *New York Times*, 8 Jan. 1958.
51. Von Braun quoted by Enid Curtis Bok Schoettle, "The Establishment of NASA," in Sanford A. Lakoff, *Knowledge and Power. Essays on Science and Government* (New York, 1966), p. 185.
52. Memo, Webb for Horace Busby (Special Assistant to the President), 23 Mar. 1965, p. 3, LBJ Library.
53. *Ibid.* Italics in original.
54. *Ibid.*, p. 2.
55. Walter Prescott Webb quoted by James Webb in draft of a speech to be delivered at Cornell University, 3 Mar. 1965, manuscript in LBJ Library. See also James E. Webb, "Education in the Space Age," *Educational Review* (Winter 1964): 33-40 (on pp. 38-39 Webb cites Frederick Jackson Turner, *The Frontier in American History* [New York, 1921], p. 283, on the link between the frontier, universities, and science). On the other frontier theorist, "the sage of West Texas," see Necah Stewart Furman, *Walter Prescott Webb. His Life and Impact* (Albuquerque, 1976).

Chapter 19

1. Editorial from *Territorial Enterprise and Virginia City News*, Nevada, Jan. 1960, copy in NASA HO.
2. Testimony of Dr. Philip Abelson, in U.S. Senate, Committee on Aeronautical and Space Sciences, *Scientists' Testimony on Space Goals*, 88th Cong., 1st Sess., 1963, pp. 3-24. Quote from p. 16.
3. *Ibid.*, pp. 62-73, 244-46.
4. Donald A. Strickland, "Physicists' Views of Space Politics," NASA General Grant NSG-243-62 (1962), NASA HO.
5. Testimony of Dr. Lloyd Berkner and Dr. Frederick Seitz, in U.S. Senate, *Scientists' Testimony on Space Goals*, pp. 88-137. Quotes from p. 89.
6. Walter Lippmann, "Money and the Moon," *Washington Post*, 2 Apr. 1962.